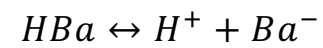


Carbonate Chemistry

- General acid / base chemistry
- Carbonate speciation
- Carbonate saturation
- Response of ocean to different scenarios

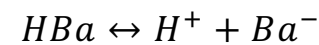
Acids and Bases in SW

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Acids and Bases in SW

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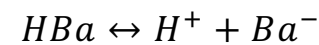
$$K' = \frac{[H^+]x[Ba^-]}{[HBa]} \quad (1)$$

Note: for carbonate chemistry we will use K' – “apparent” equilibrium constants.

- These are determined experimentally using near-seawater concentrations of ions
- Include impact of activity w/o additional calculation

Acids and Bases in SW

- Dissolving species HBa in water gives H^+ and Ba^-
- Balance between species described by equilibrium constant: K'
- Total amount of Ba in solution is the combination of $[HBa]$ and $[Ba^-]$

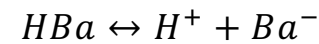


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Acids and Bases in SW

- Dissolving species HBa in water gives H^+ and Ba^-
- Balance between species described by equilibrium constant: K'
- Total amount of Ba in solution is the combination of $[HBa]$ and $[Ba^-]$
- Seawater is neutral, so charge balance must be maintained

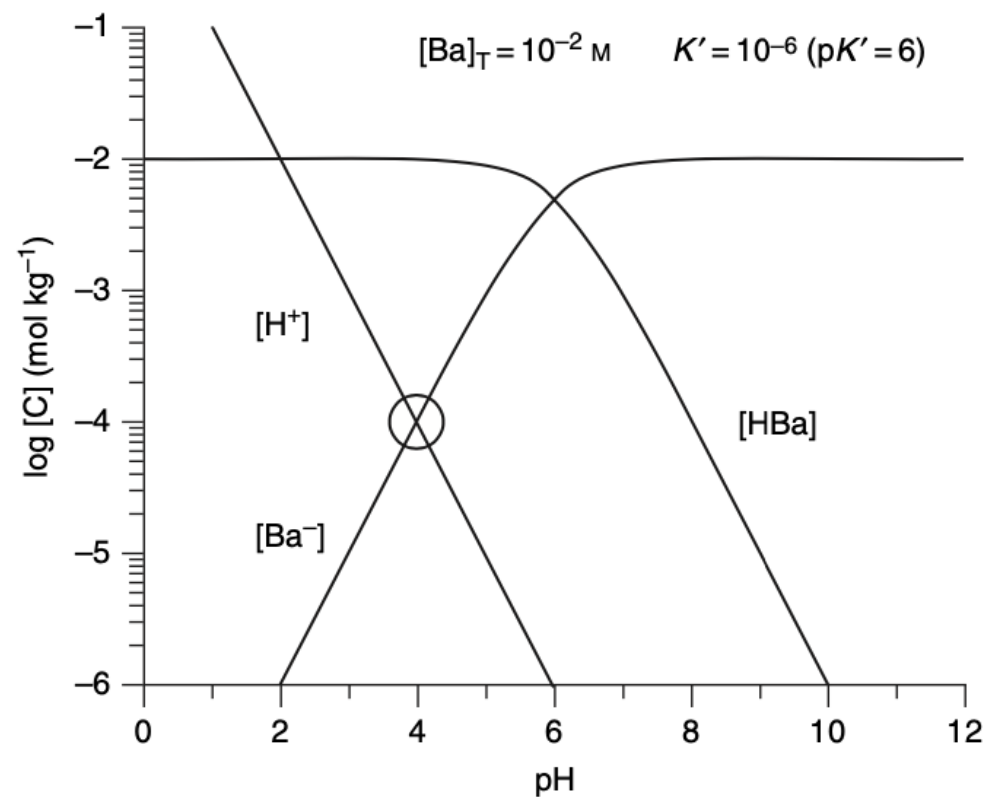


$$K' = \frac{[H^+][Ba^-]}{[HBa]} \quad (1)$$

$$[Ba]_T = [HBa] + [Ba^-] \quad (2)$$

$$0 = [H^+] - [Ba^-] \quad (3)$$

Acid-Bases: Log species distribution vs. pH



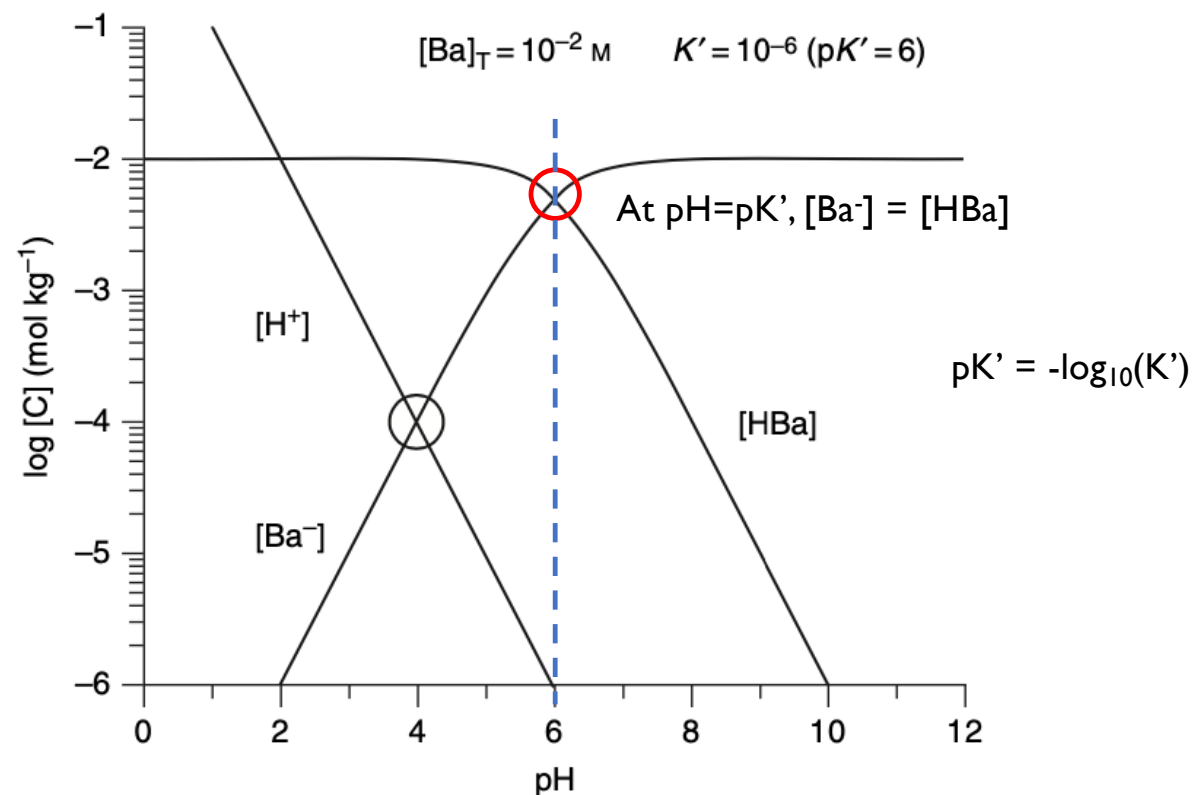
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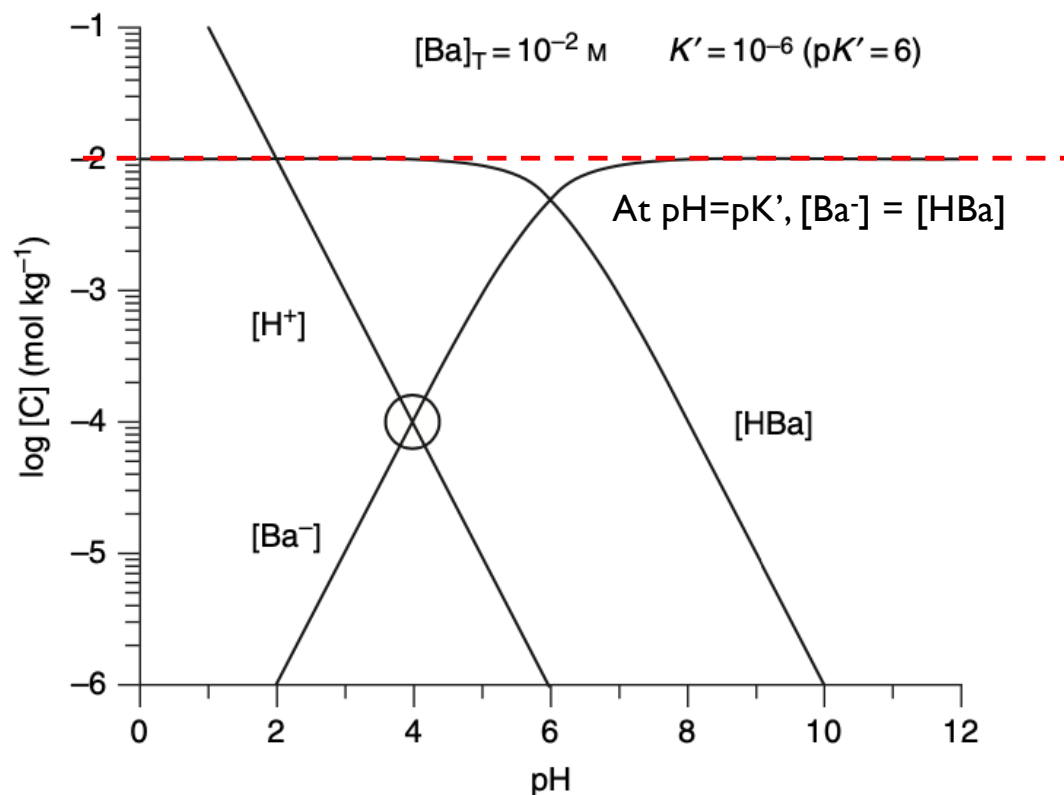
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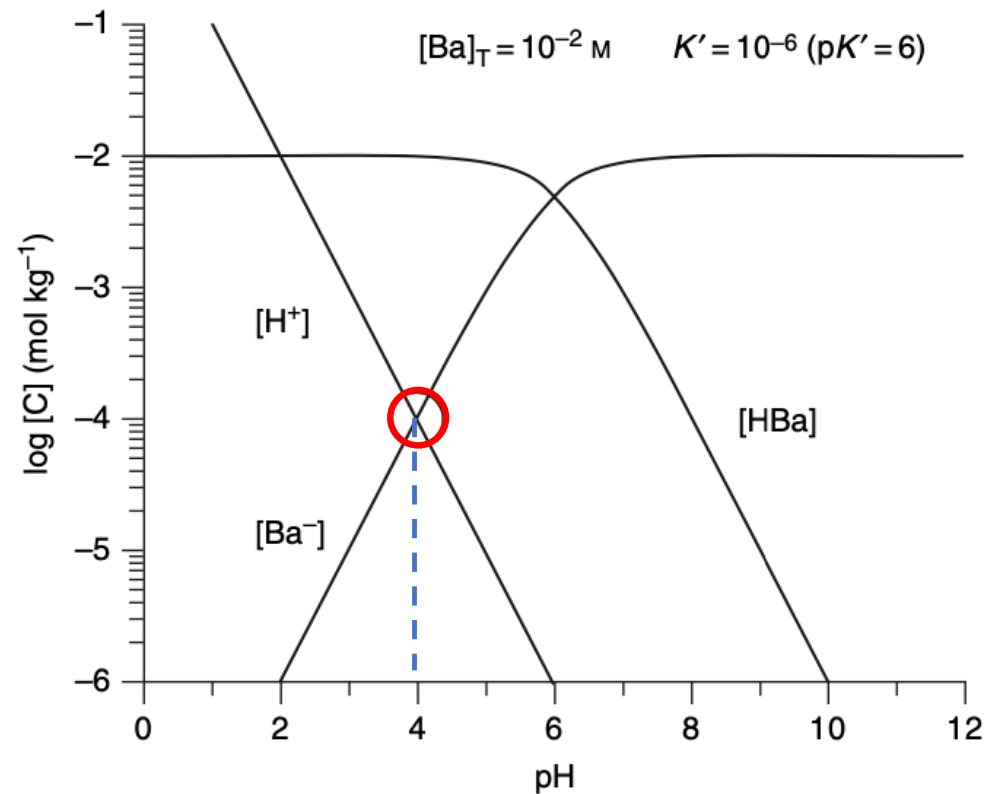
Acid-Bases: Log species distribution vs. pH



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- pH of this system will equal 4

Acid-Bases: Log species distribution vs. pH

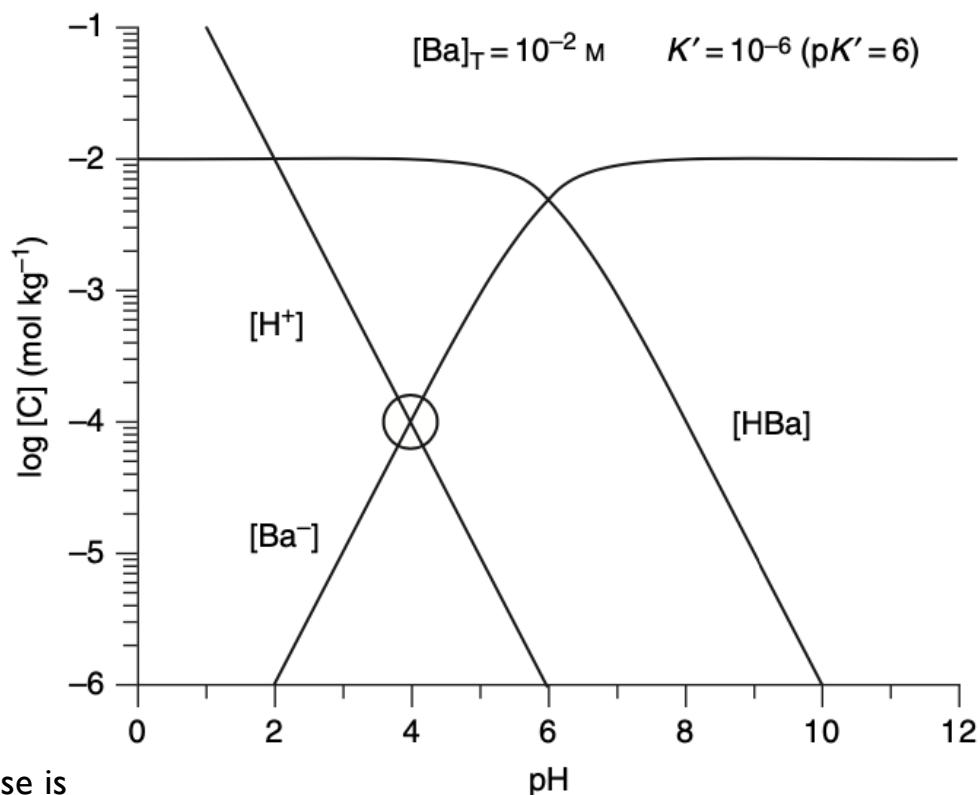


$$K' = \frac{[H^+][Ba^-]}{[HBa]} \quad (1)$$

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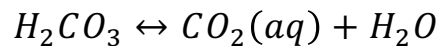
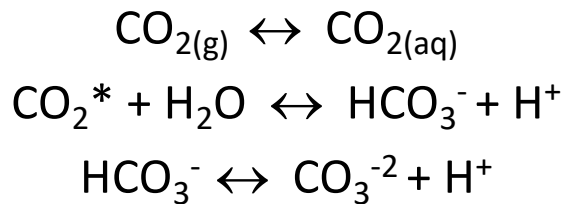
- 3 equations, 4 unknowns (H, Ba, HBa, $[Ba]_T$)
- Charge balance sets pH, but don't know concentrations of species unless something else is measured ($[Ba]_T$)



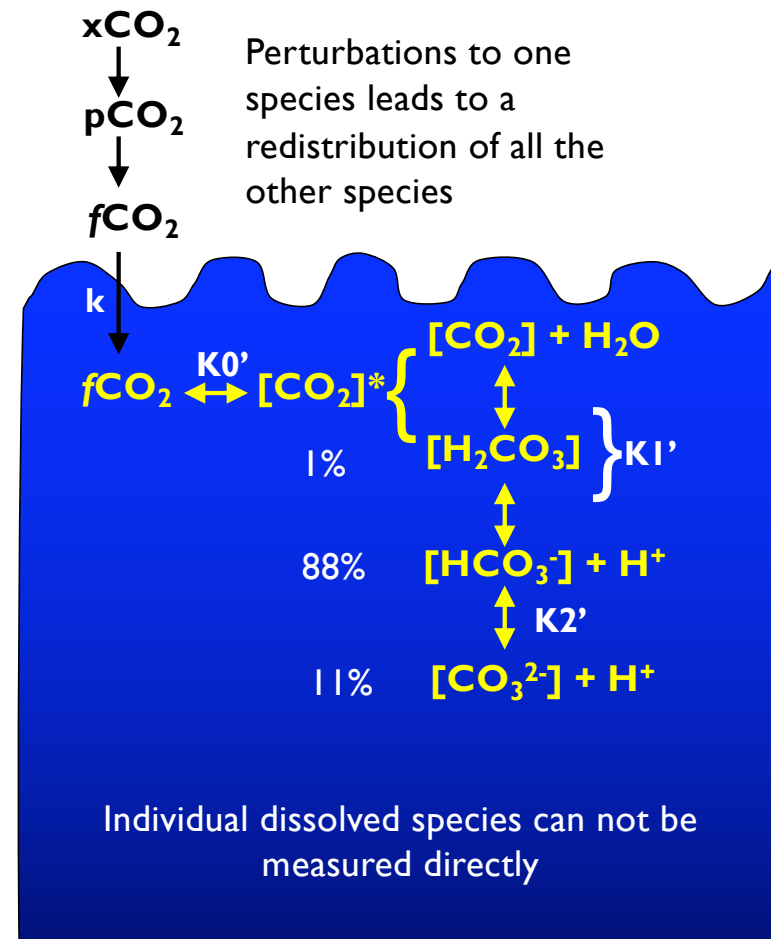
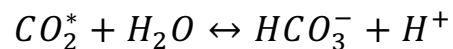
Why do we study the CO₂ system?

- CO₂ controls the fraction of inbound radiation that remains trapped in the atmosphere (**greenhouse effect**), which in turn strongly influences planetary climate
- CO₂ is the raw material used to build **organic matter**
- CO₂ controls the **pH** of the oceans
- Distribution of CO₂ species affects **preservation of CaCO₃** deposited on the sea floor

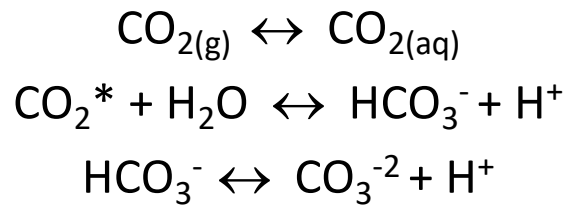
CO₂ Speciation



- Dissolved CO₂ hydrates to form carbonic acid (H₂CO₃) which then dissociates to form HCO₃⁻
- In practice, very little dissolved CO₂ exists as carbonic acid and it is not possible to distinguish between carbonic acid and CO₂(aq), so a composite equation is used:

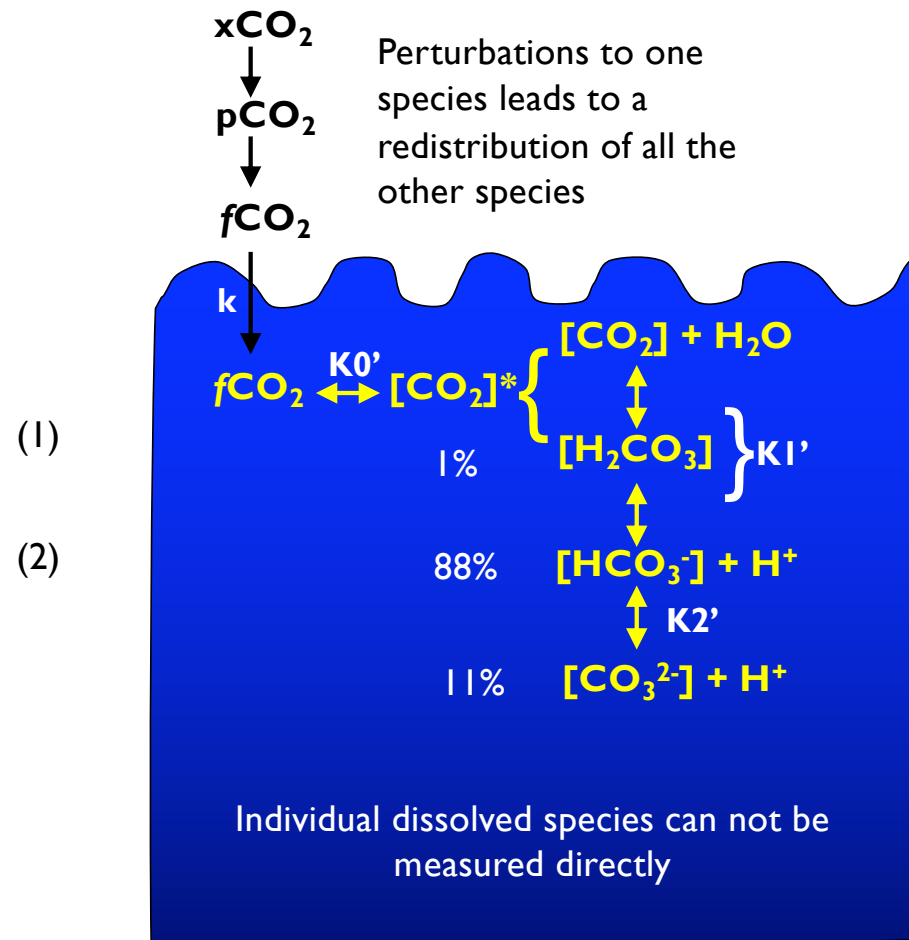


CO₂ Speciation

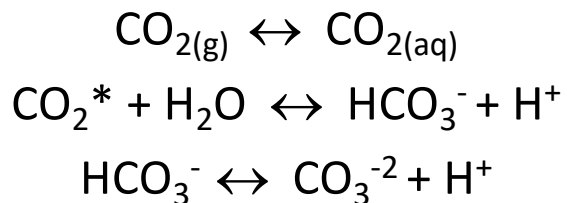


$$K'_1 = \frac{\{\text{H}^+\}[\text{HCO}_3^-]}{[\text{CO}_2]^*}$$

$$K'_2 = \frac{\{\text{H}^+\}[\text{CO}_3^{2-}]}{[\text{HCO}_3^-]}$$



CO₂ Speciation



$$K'_1 = \frac{\{\text{H}^+\}[\text{HCO}_3^-]}{[\text{CO}_2]^*}$$

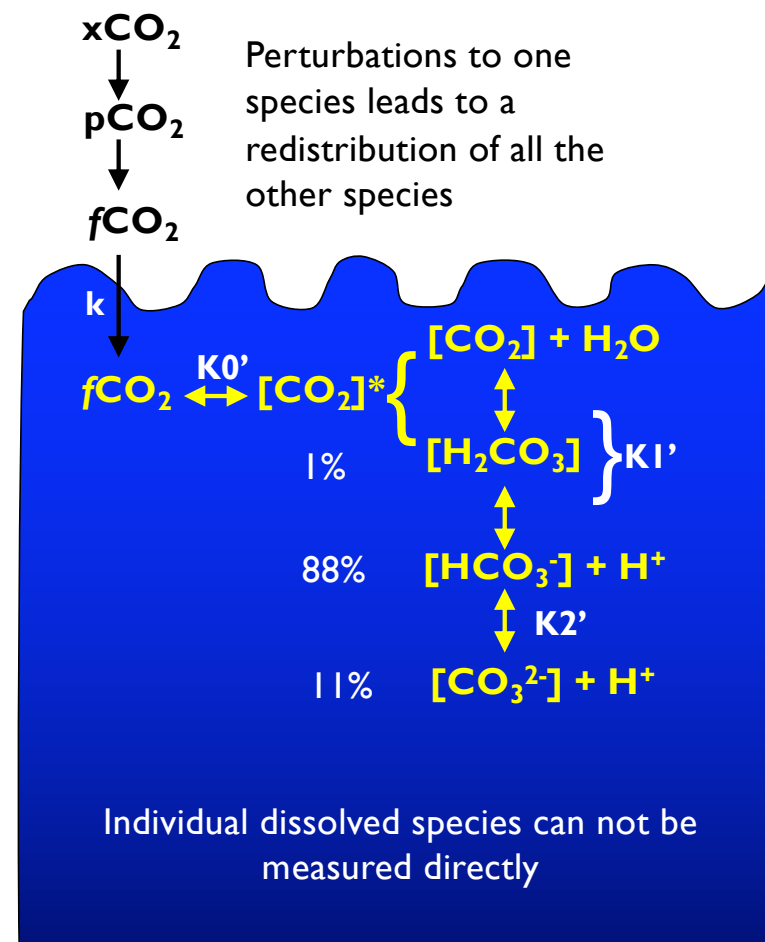
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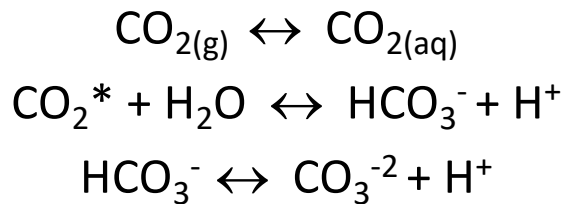
(1)

(2)

(3)



CO₂ Speciation



$$K'_1 = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{CO}_2]^*}$$

(1)

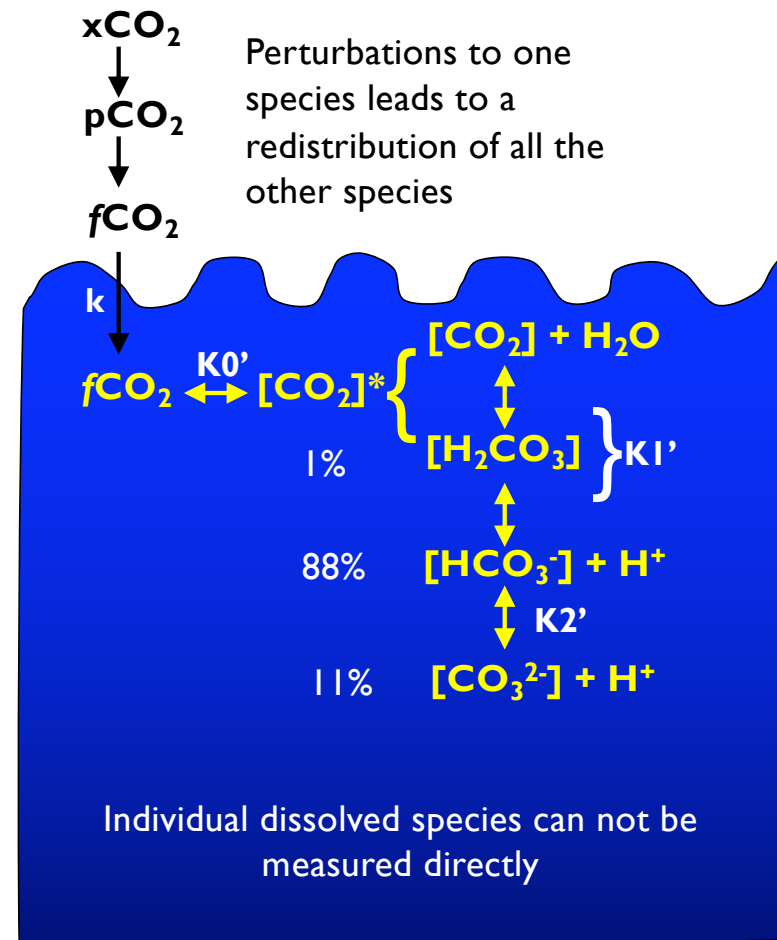
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(2)

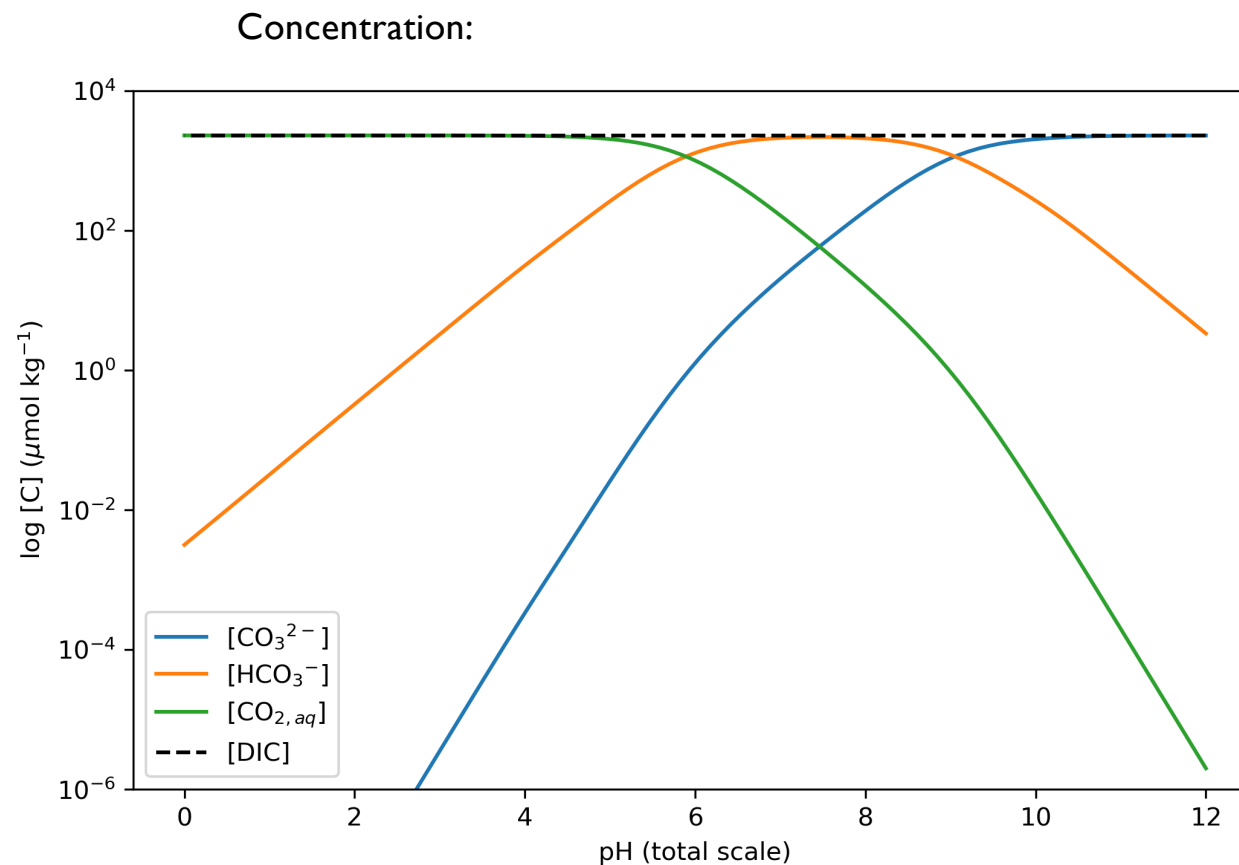
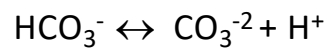
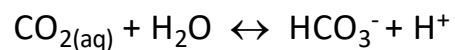
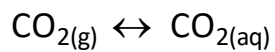
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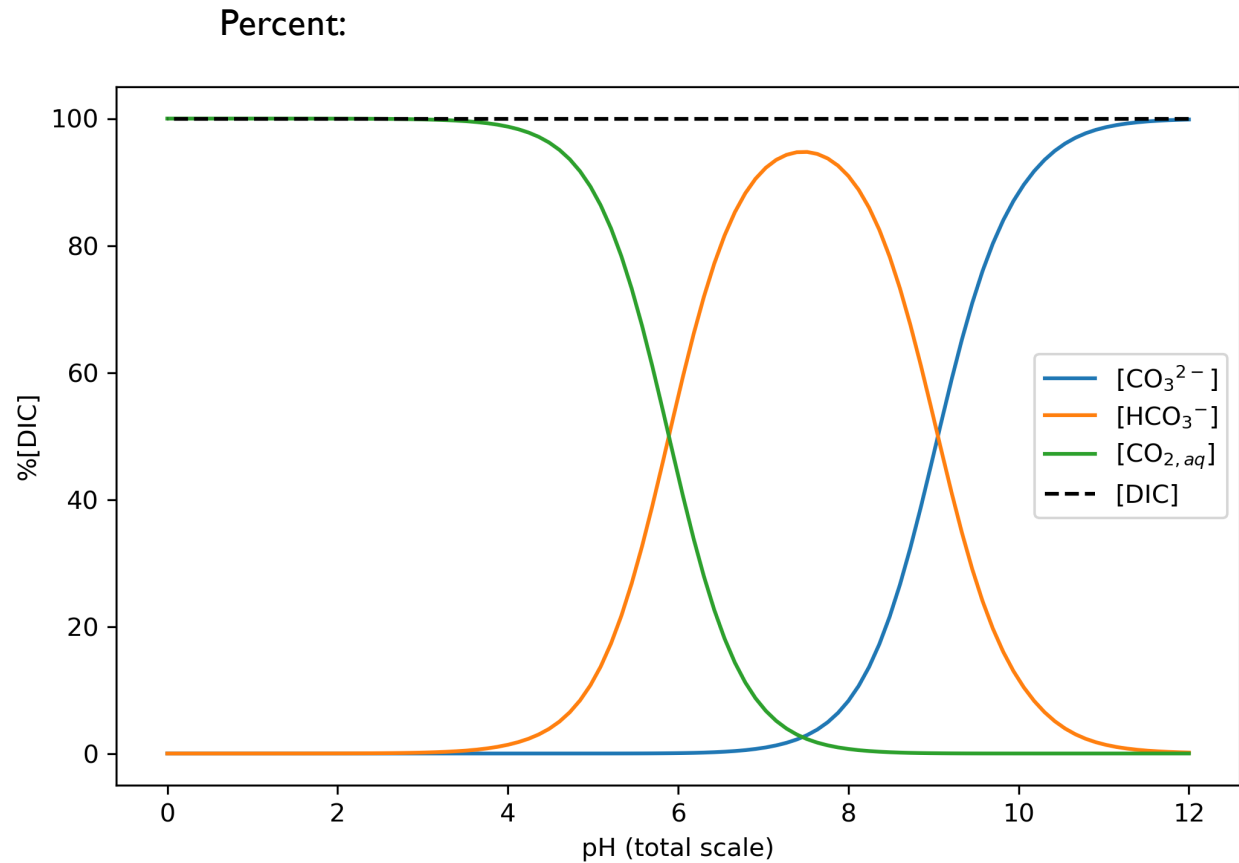
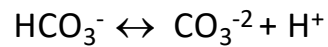
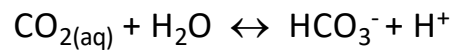
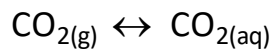
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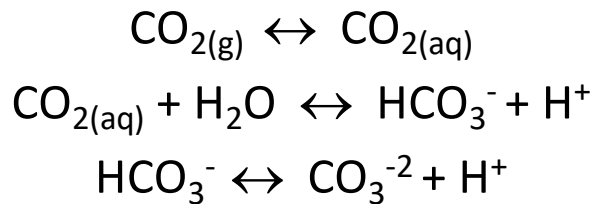
Distribution of CO₂ Species at Different pH Values



Distribution of CO₂ Species at Different pH Values



CO₂ Speciation



- What happens when we start perturbing the system?
- Addition of CO₂?
 - DIC? TA? pH?

$$K'_1 = \frac{\{\text{H}^+\}[\text{HCO}_3^-]}{[\text{CO}_2]^*} \quad (1)$$

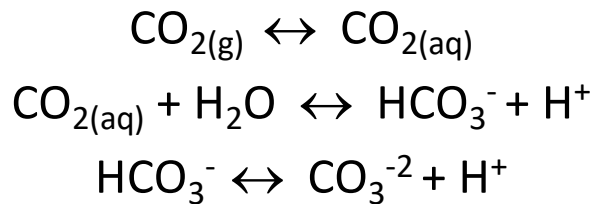
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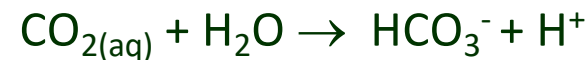
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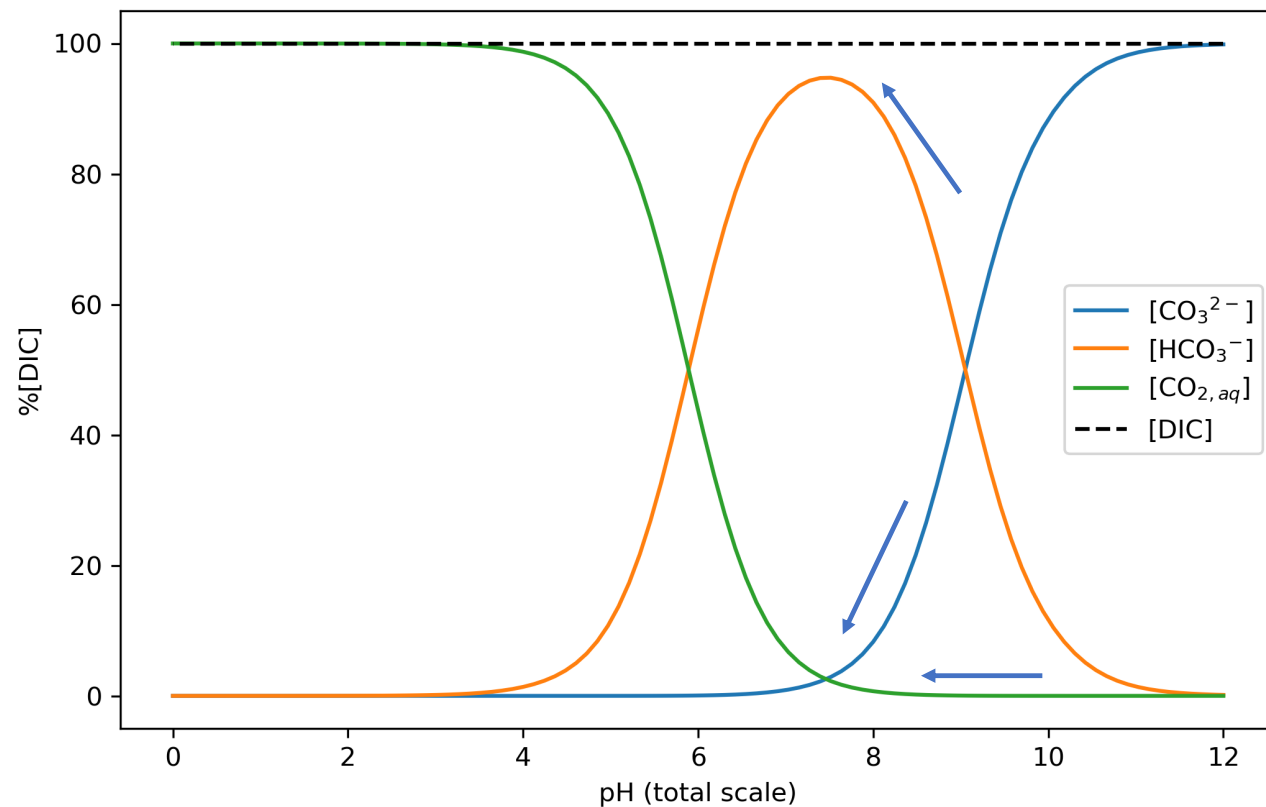
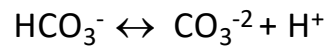
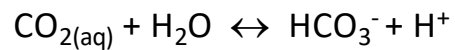
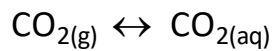
- What happens when we start perturbing the system?
 - Addition of CO₂?



The limited amount of CO₃²⁻ available means that not all of the H⁺ produced by the middle reaction can be consumed

- What about impact on alkalinity?

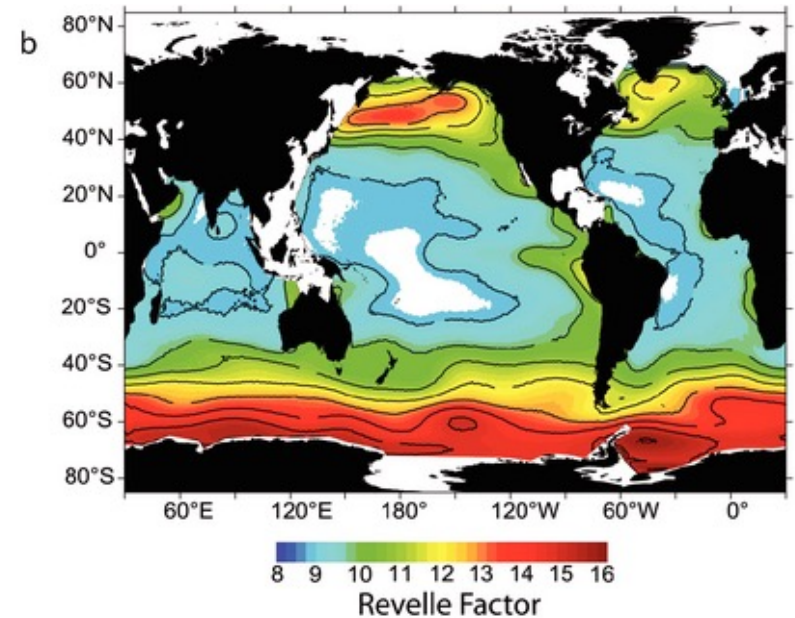
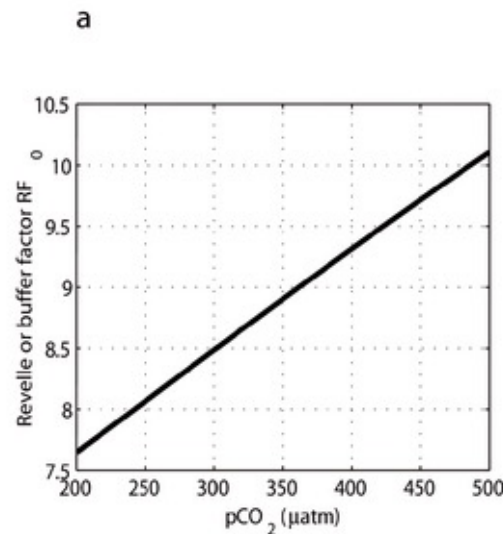
Distribution of CO₂ Species at Different pH Values



Revelle factor – buffer capacity of seawater

- How much will a given change in carbon content impact seawater $p\text{CO}_2$?

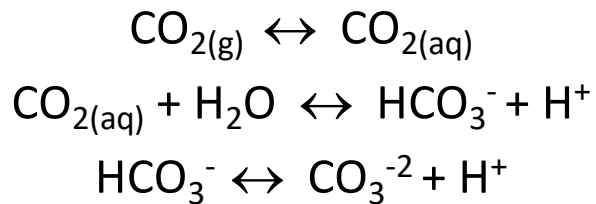
$$R_{Rev} = \frac{\Delta f\text{CO}_2 / f\text{CO}_2}{\Delta \text{DIC} / \text{DIC}}$$



IPCC AR4, Fig. 7-11

v2025

CO₂ Speciation



- What happens when we start perturbing the system?
 - Addition of CO₂?
 - Precipitation of CaCO₃?

$$K'_1 = \frac{\{\text{H}^+\}[\text{HCO}_3^-]}{[\text{CO}_2]^*} \quad (1)$$

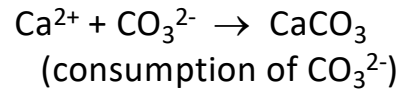
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- 4 equations, 6 unknowns (H⁺, HCO₃⁻, CO₂, CO₃²⁻, DIC, TA)

Precipitation of CaCO_3 ?



Does this reduce the CO_2 (pCO_2)
level of the seawater?

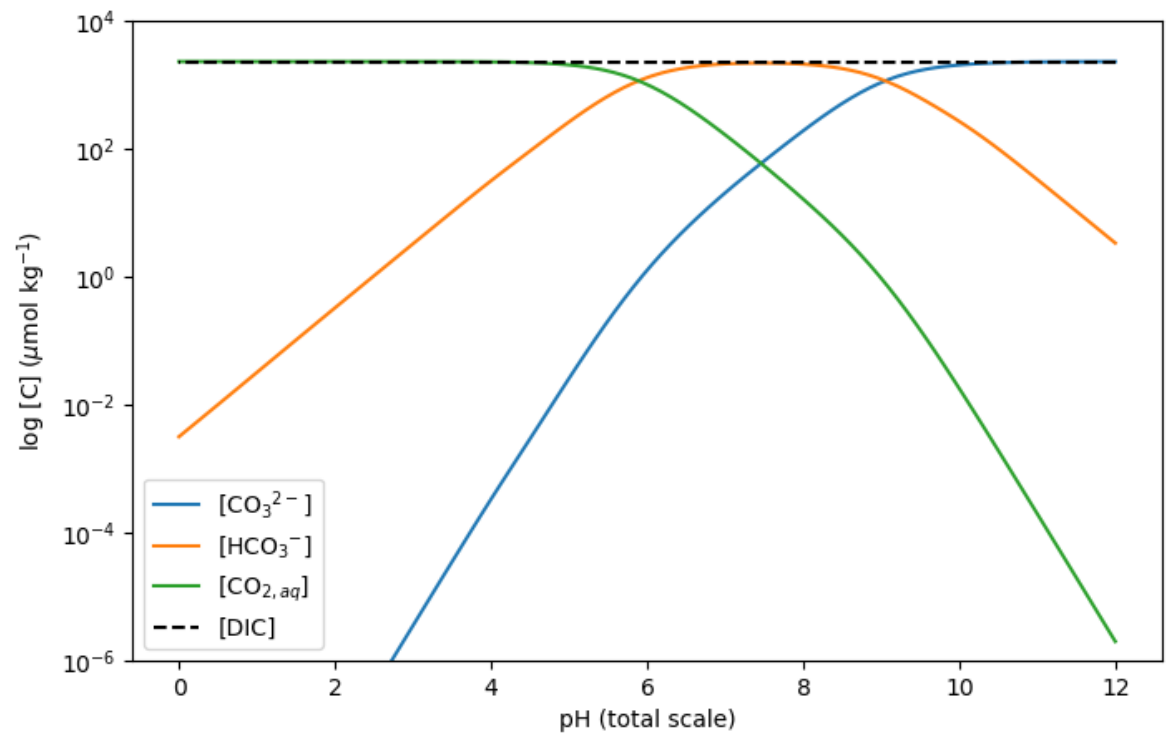
No! Lost CO_3^{2-} will be replaced:



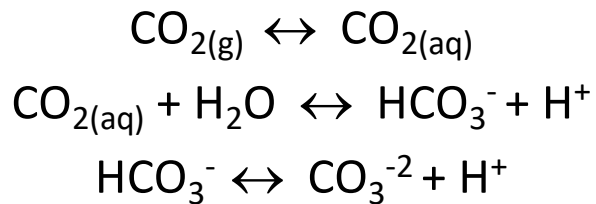
But this H^+ release causes:



Thus, CaCO_3 precipitation causes
a decrease in DIC and TA, but
an increase in pCO_2



CO₂ Speciation



- What happens when we start perturbing the system?
 - Addition of CO₂?
 - Precipitation of CaCO₃?
 - Formation of organic matter?

$$K'_1 = \frac{\{\text{H}^+\}[\text{HCO}_3^-]}{[\text{CO}_2]^*} \quad (1)$$

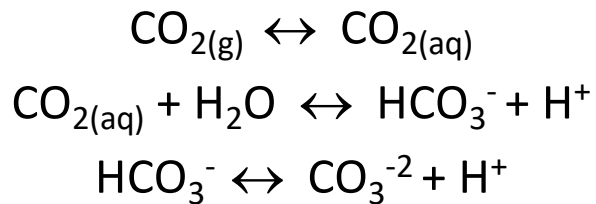
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CO₂ Speciation



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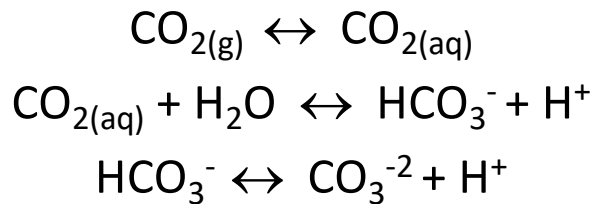
- What happens when we start perturbing the system?
 - Addition of CO₂?
 - Precipitation of CaCO₃?
 - Formation of organic matter?



$$(1) \quad \text{Usage of P:N:C} = 1:16:106$$

$$(2) \quad \text{1 mol of OM formed} = \text{decrease of DIC by 1}$$

CO₂ Speciation



$$K'_1 = \frac{\{\text{H}^+\}[\text{HCO}_3^-]}{[\text{CO}_2]^*}$$

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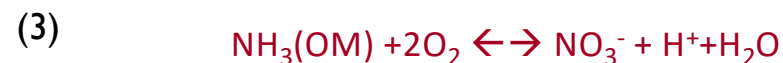
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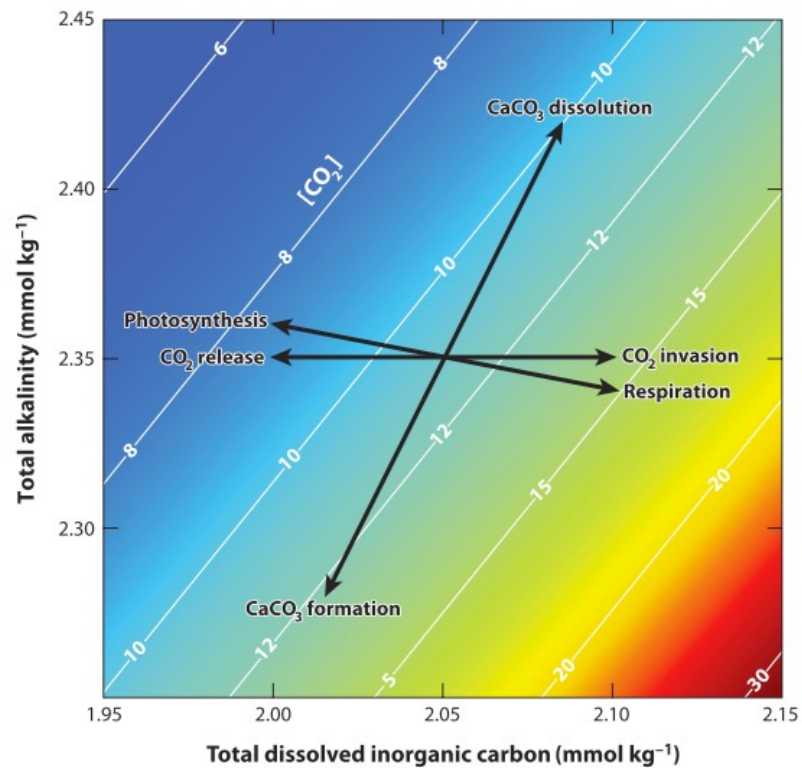
(1) Usage of P:N:C = 1:16:106

- (2)
- 1 mol of OM formed = decrease of DIC by 1
 - Nitrate incorporation / degradation changes H⁺



- (4)
- Changes alkalinity in a -16/106 ratio (0.15, opposite sign to DIC change)

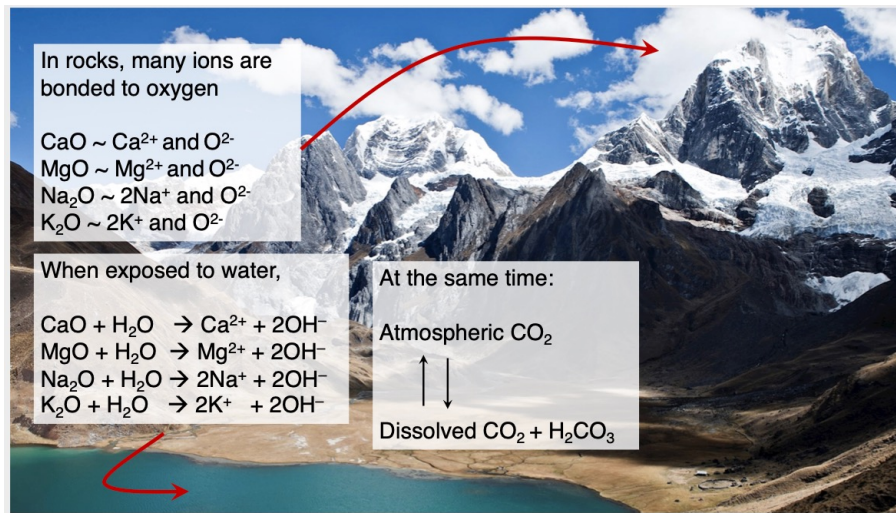
Carbonate changes



- What happens when we start perturbing the system?
- Addition of CO₂?
- Precipitation of CaCO₃?
- Formation of organic matter?

- Zeebe (2012)

Seawater carbon content vs. freshwater



In rocks, many ions are bonded to oxygen

CaO ~ Ca²⁺ and O²⁻
MgO ~ Mg²⁺ and O²⁻
Na₂O ~ 2Na⁺ and O²⁻
K₂O ~ 2K⁺ and O²⁻

When exposed to water,

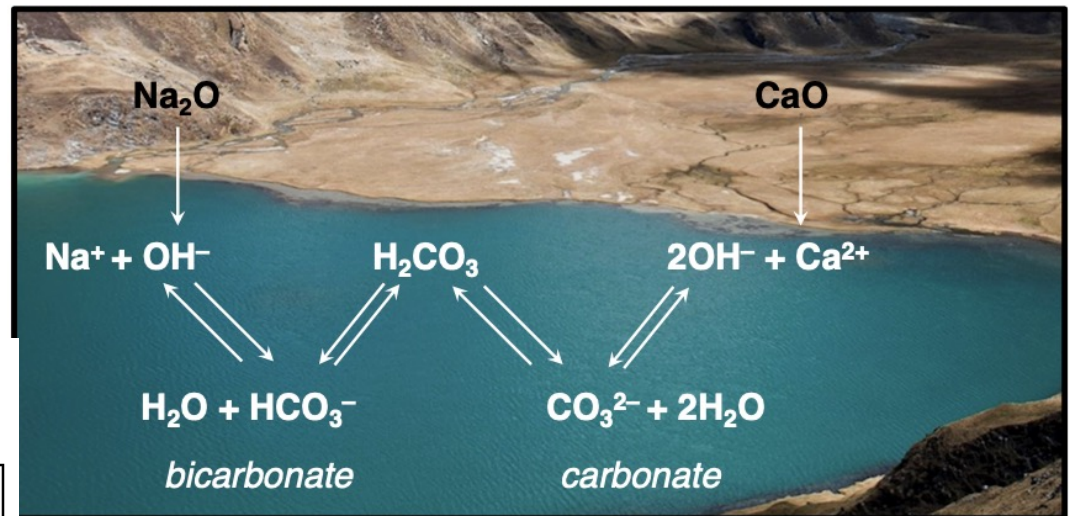
CaO + H₂O → Ca²⁺ + 2OH⁻
MgO + H₂O → Mg²⁺ + 2OH⁻
Na₂O + H₂O → 2Na⁺ + 2OH⁻
K₂O + H₂O → 2K⁺ + 2OH⁻

At the same time:

Atmospheric CO₂
↑ ↓
Dissolved CO₂ + H₂CO₃

Weathering of rocks forms HCO₃⁻ and CO₃²⁻
These additional pools of carbon allow seawater to take up much more carbon than freshwater

	Pure water	Ocean water
Dissolved CO ₂ + carbonic acid	□ 0.4 mg/L	□ 0.4 mg/L
Bicarbonate + Carbonate	▪ < 0.04 mg/L	>90 mg/L

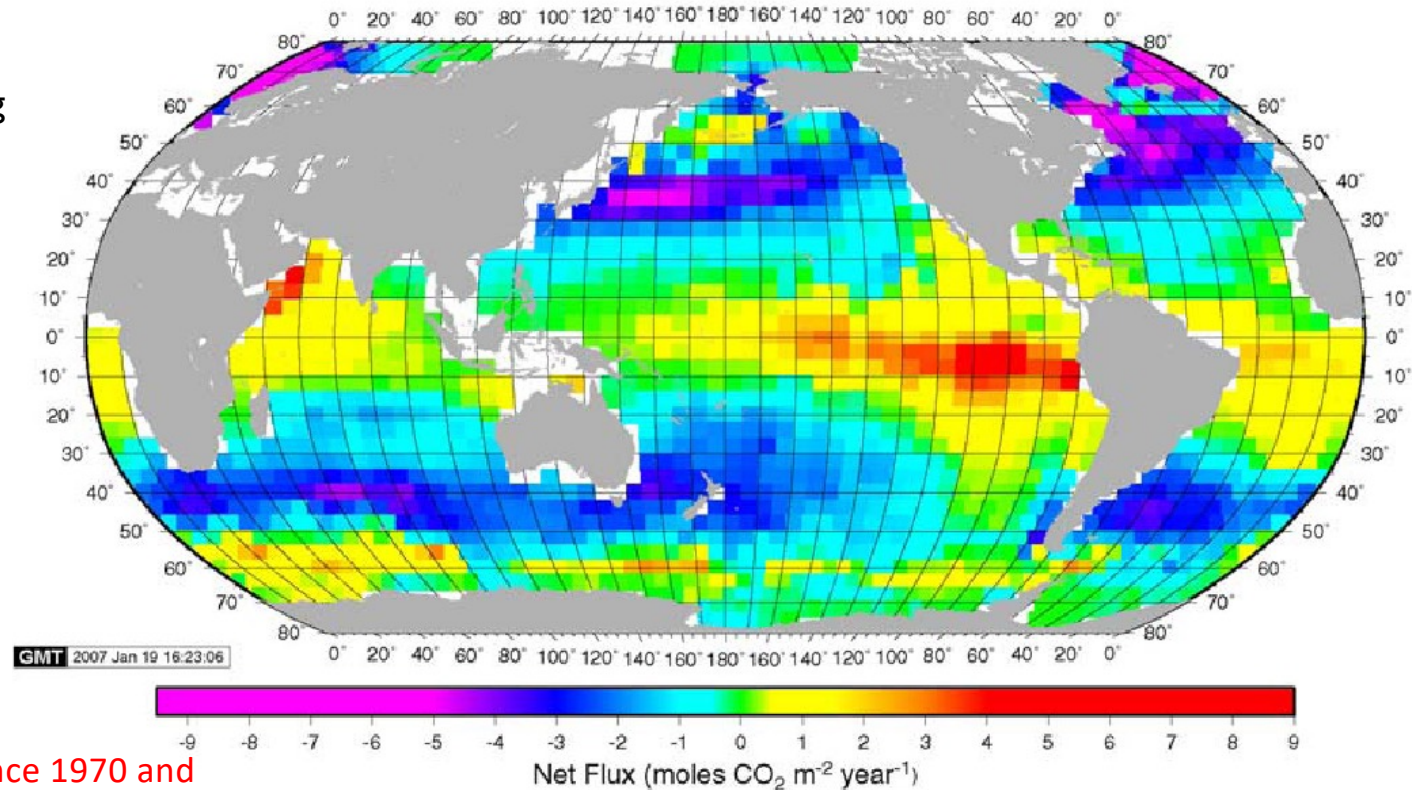


SW can hold ~ 200x more carbon than FW

CO₂ Speciation – fCO₂

Takahashi climatological annual mean air-sea CO₂ flux for reference year 2000

- fCO₂ is measured by equilibrating seawater with a headspace, then measuring the CO₂ molecules in that headspace using infrared adsorption



Based on ~3 million measurements since 1970 and
NCEP/DOE/AMIP II reanalysis.
Global flux is 1.4 ± 0.7 Pg C/yr

- (Current SOCAT database has >21 million measurements)

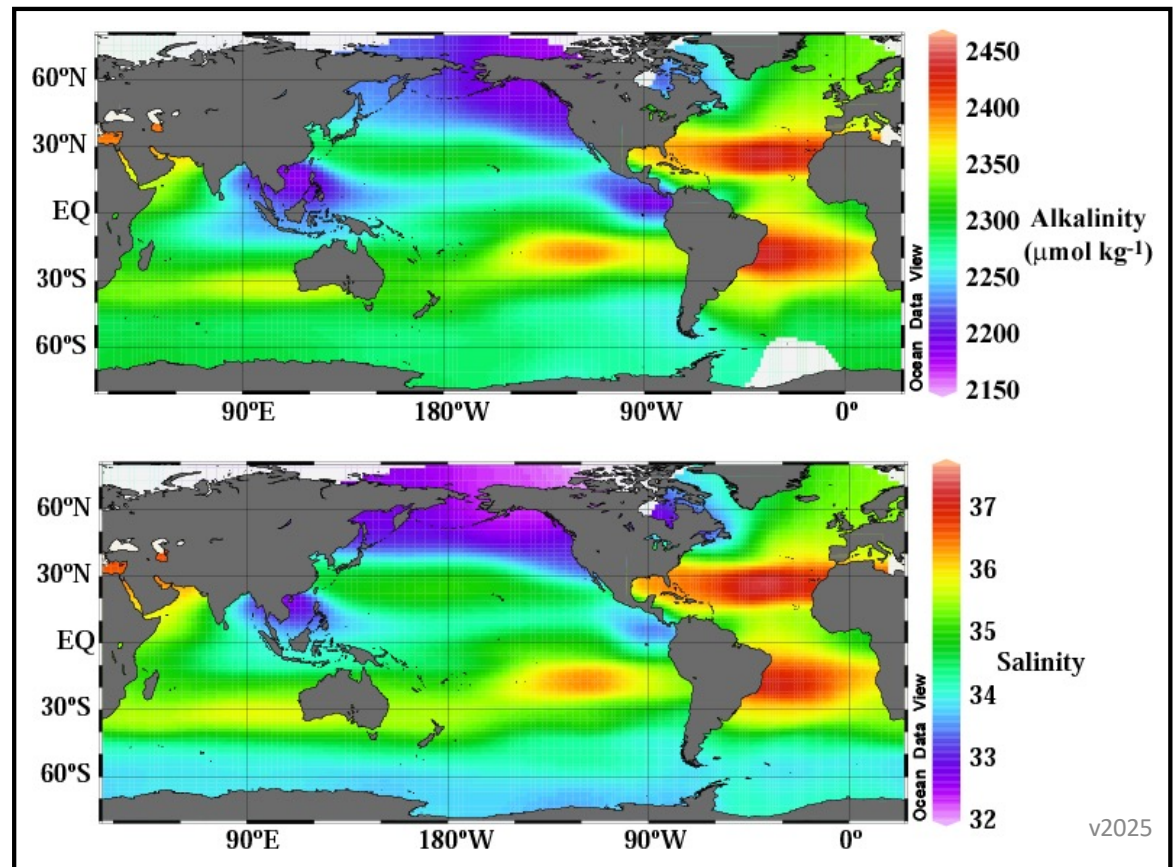
CO₂ Speciation – Total Alkalinity

- TAlk is measured by titrating SW with acid.
- While the the seawater is still being buffered, TA>0 (i.e. there are smaller changes in measured H⁺ changes than expected)
- Once measured H⁺ change equals observed H⁺ change, you know that all weak acids are gone, so TA can be calculated based on the amount of acid added

$$TA = [HCO_3^-] + 2*[CO_3^{2-}] + [B(OH)_4^-] + [HPO_4^{2-}] + 2*[PO_4^{3-}]... + [OH^-] - [H^+]$$

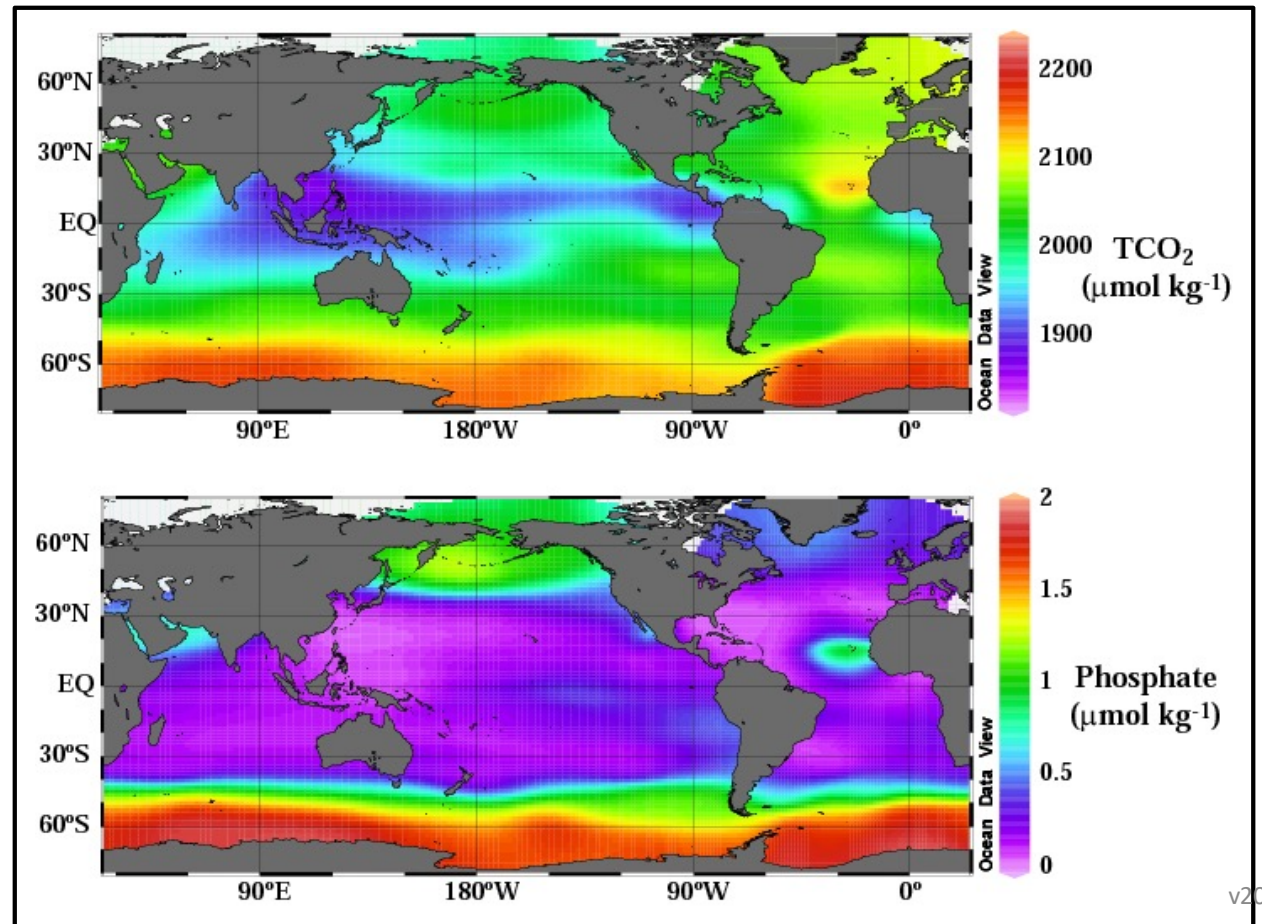
Carbonate alkalinity – simplification for calculation in Psets: $A_c = [HCO_3^-] + 2[CO_3^{2-}]$

Surface TA/TAlk/Alk/A_T Distribution is Very Similar to Salinity



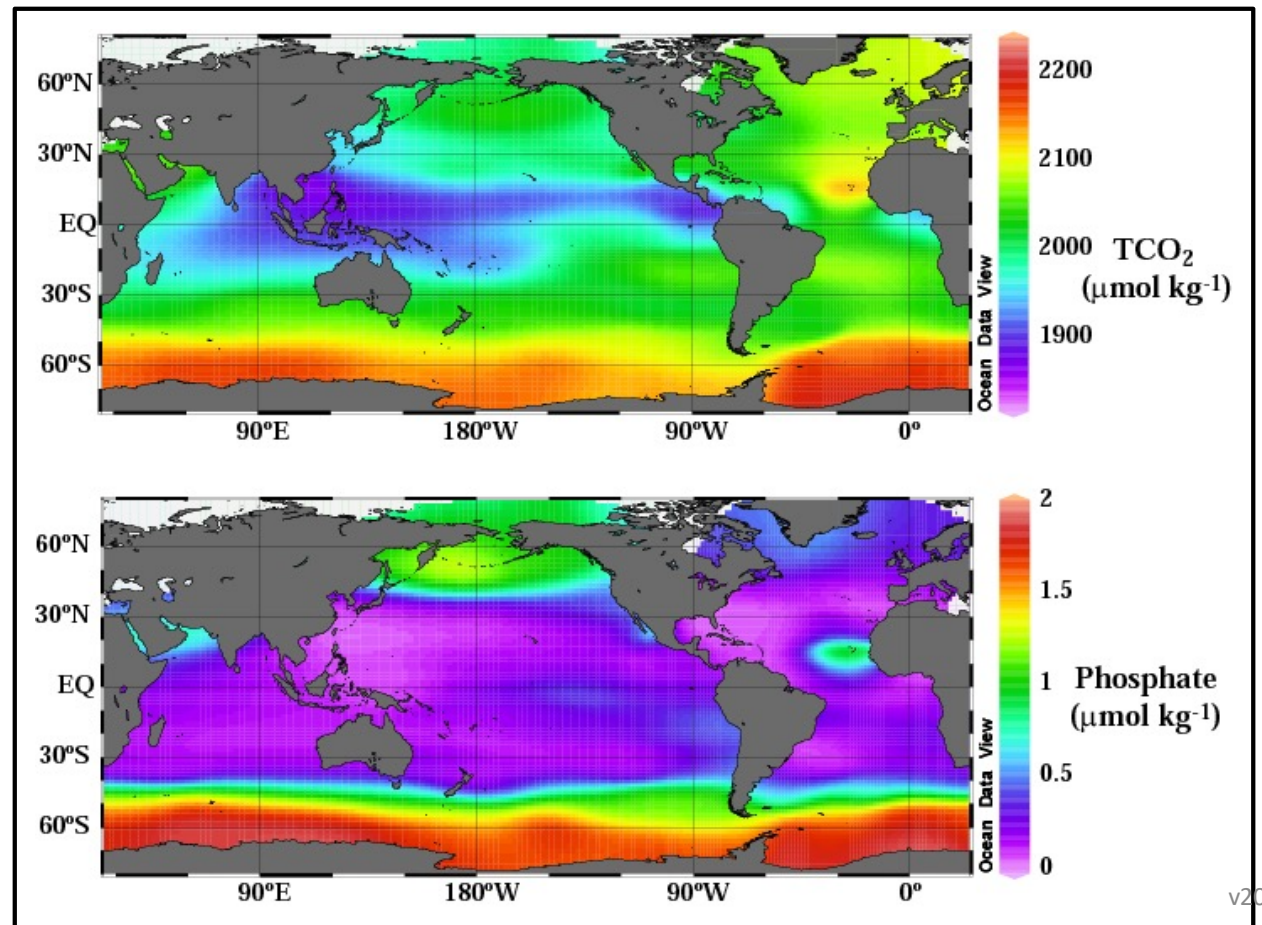
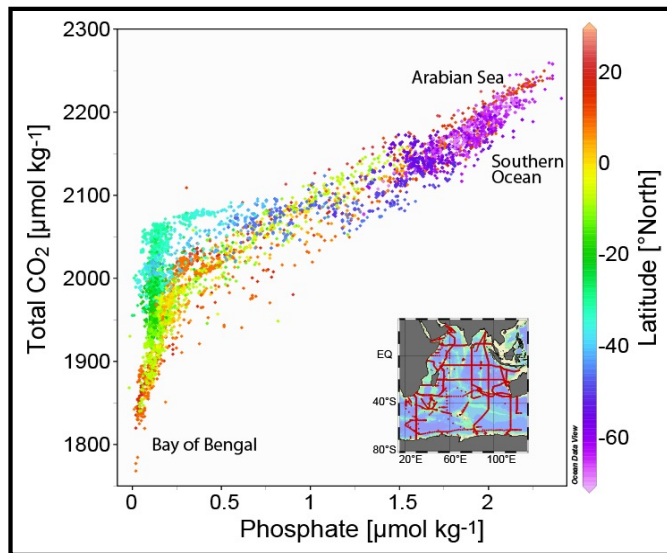
CO₂ Speciation – Total CO₂

TCO₂/DIC/C_T: Surface Distribution
is Similar to Nutrient Distributions



CO₂ Speciation – Total CO₂

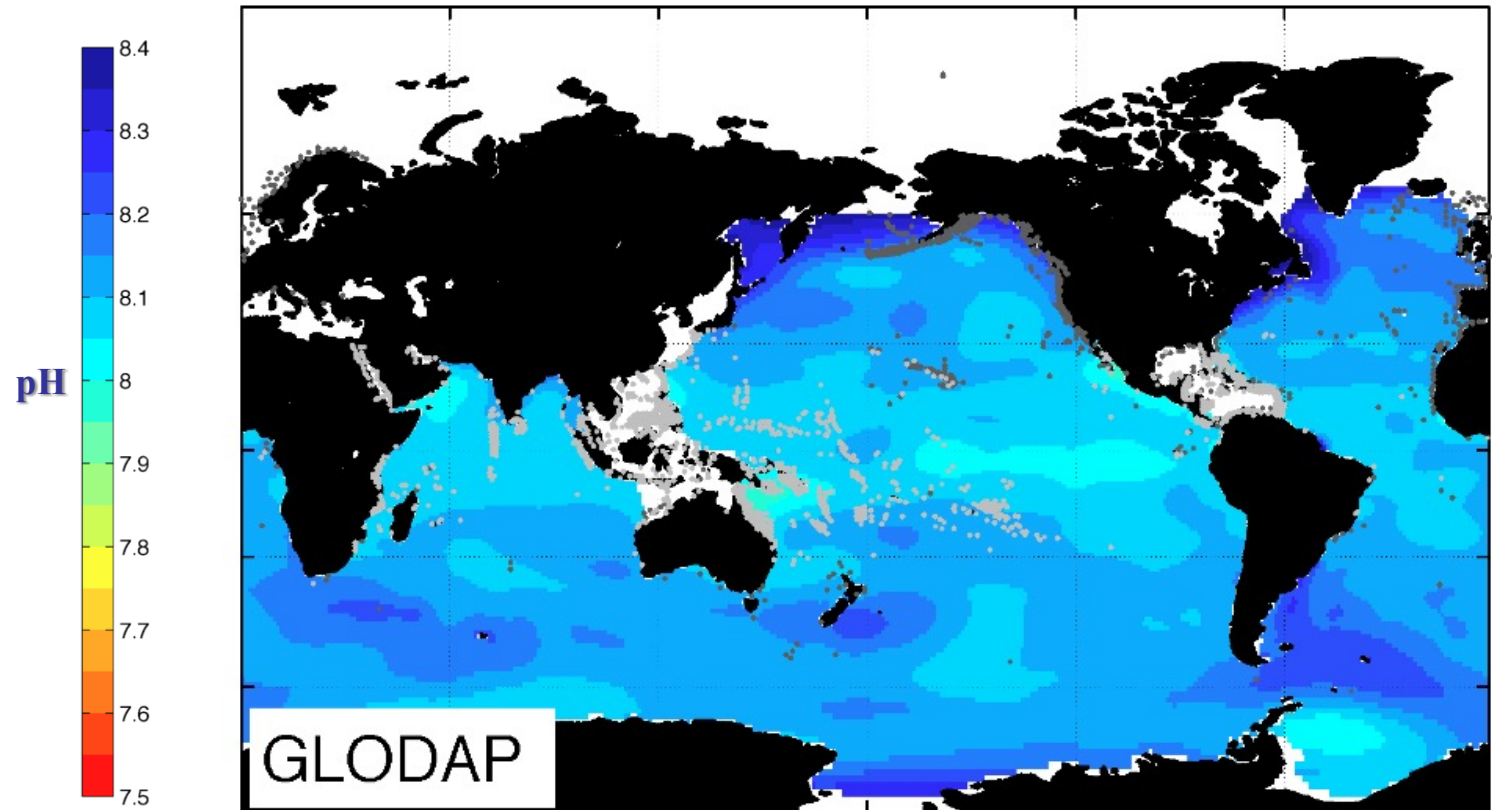
Shallow Indian Ocean Data
(depth < 100 m)



CO₂ Speciation – pH

Surface pH distribution reflects combined patterns of alkalinity and TCO₂

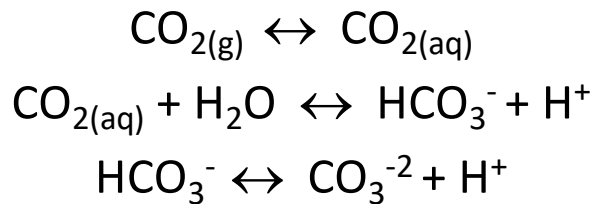
- Measured on several different scales (free, Total, seawater)
- Important to be consistent
 - Total scale (pH_T) most commonly used in oceanography



Light gray = warm water corals
Dark gray = cold water corals

Feely, Doney and Cooley, Oceanography (2009)

CO₂ Speciation



$$K'_1 = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{CO}_2]^*}$$

(1)

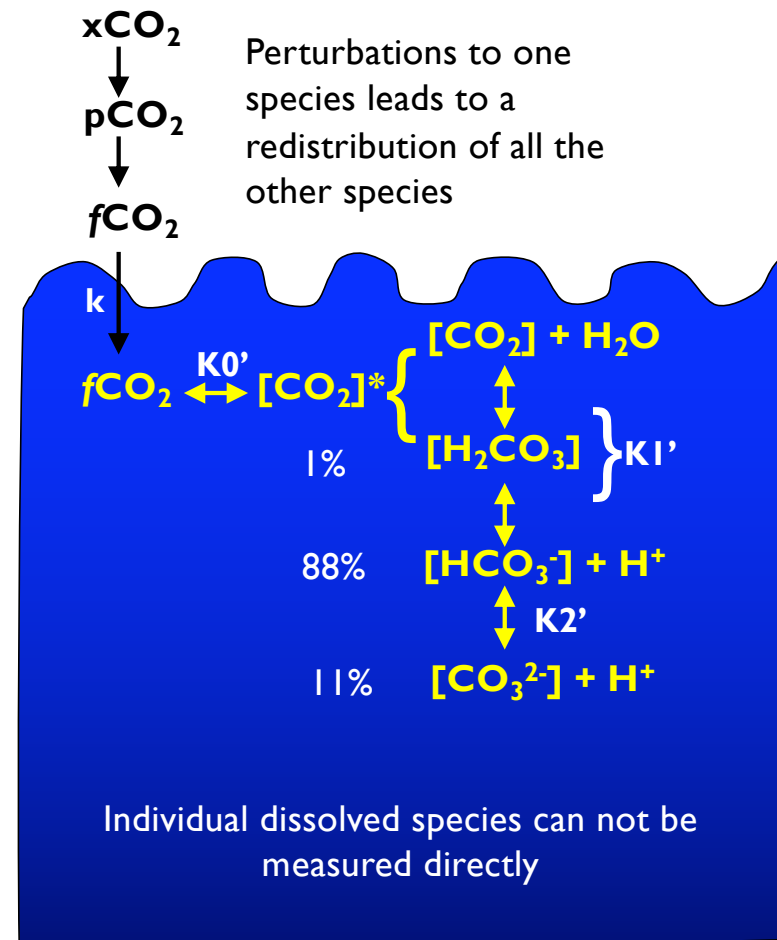
$$K'_2 = \frac{[\text{H}^+][\text{CO}_3^{2-}]}{[\text{HCO}_3^-]}$$

(2)

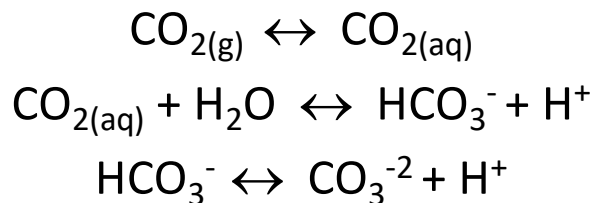
$$\text{DIC (or TCO}_2) = [\text{CO}_2]^* + [\text{HCO}_3^-] + [\text{CO}_3^{2-}] \quad (3)$$

$$\text{TA} = [\text{HCO}_3^-] + 2*[\text{CO}_3^{2-}] + [\text{B(OH)}_4^-] + [\text{HPO}_4^{2-}] + 2*[\text{PO}_4^{3-}] \dots + [\text{OH}^-] - [\text{H}^+] \quad (4)$$

- 4 equations, 6 unknowns (H⁺, HCO₃⁻, CO₂, CO₃²⁻, DIC, TA)
- Measuring 2 allows the system to be constrained



CO₂ Speciation – For HW



$$K'_1 = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{CO}_2]^*}$$

(1)

$$K'_2 = \frac{[\text{H}^+][\text{CO}_3^{2-}]}{[\text{HCO}_3^-]}$$

(2)

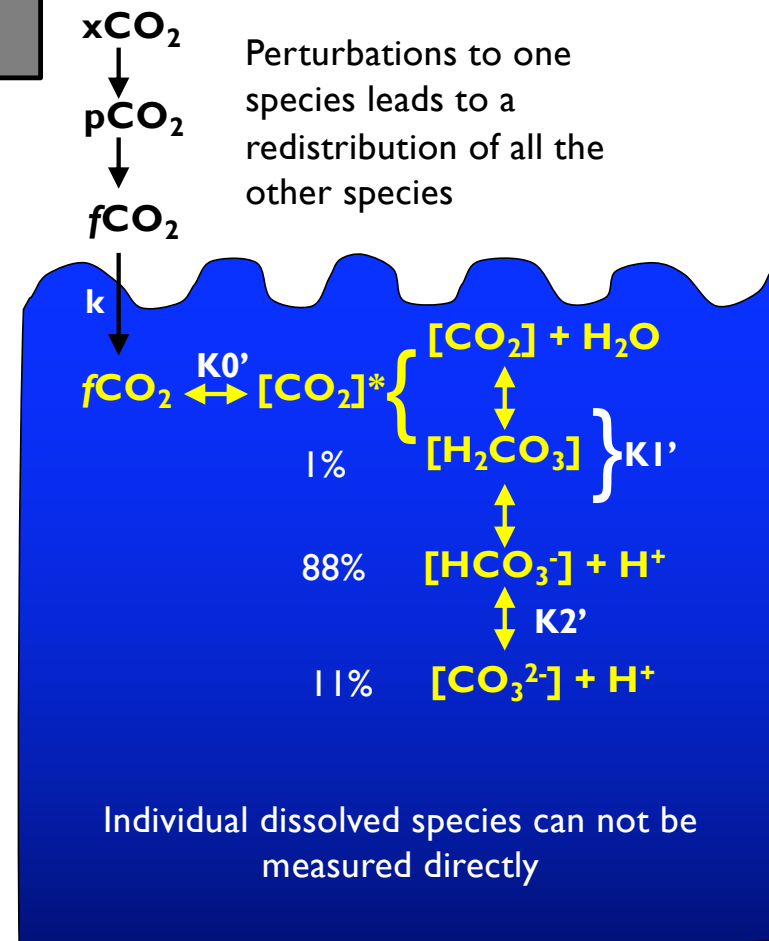
$$\text{DIC (or TCO}_2) = [\text{HCO}_3^-] + [\text{CO}_3^{2-}]$$

(3)

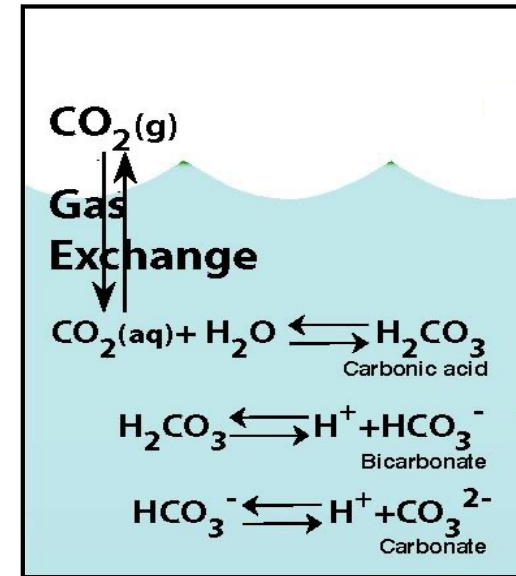
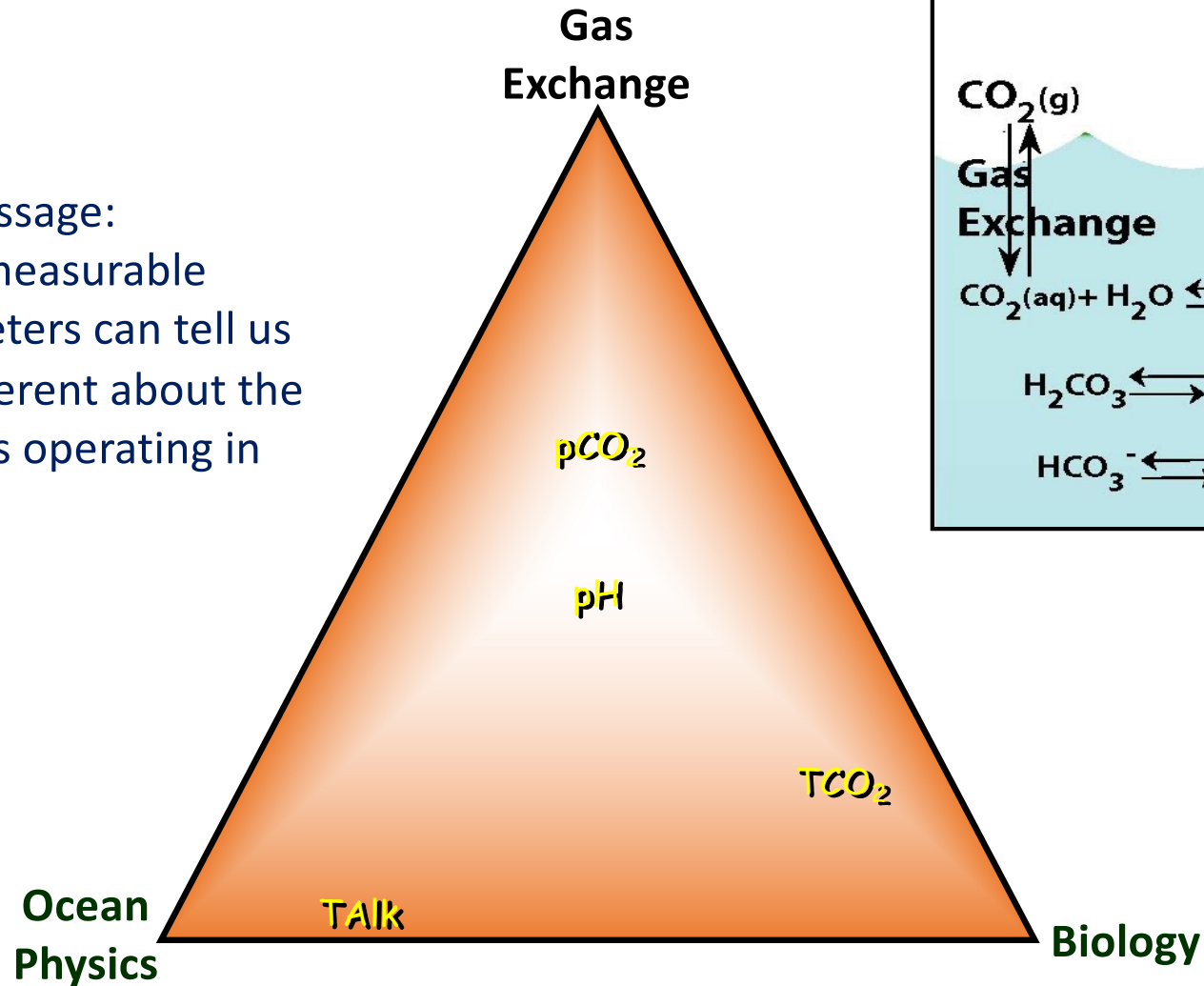
$$A_c = [\text{HCO}_3^-] + 2[\text{CO}_3^{2-}]$$

(4)

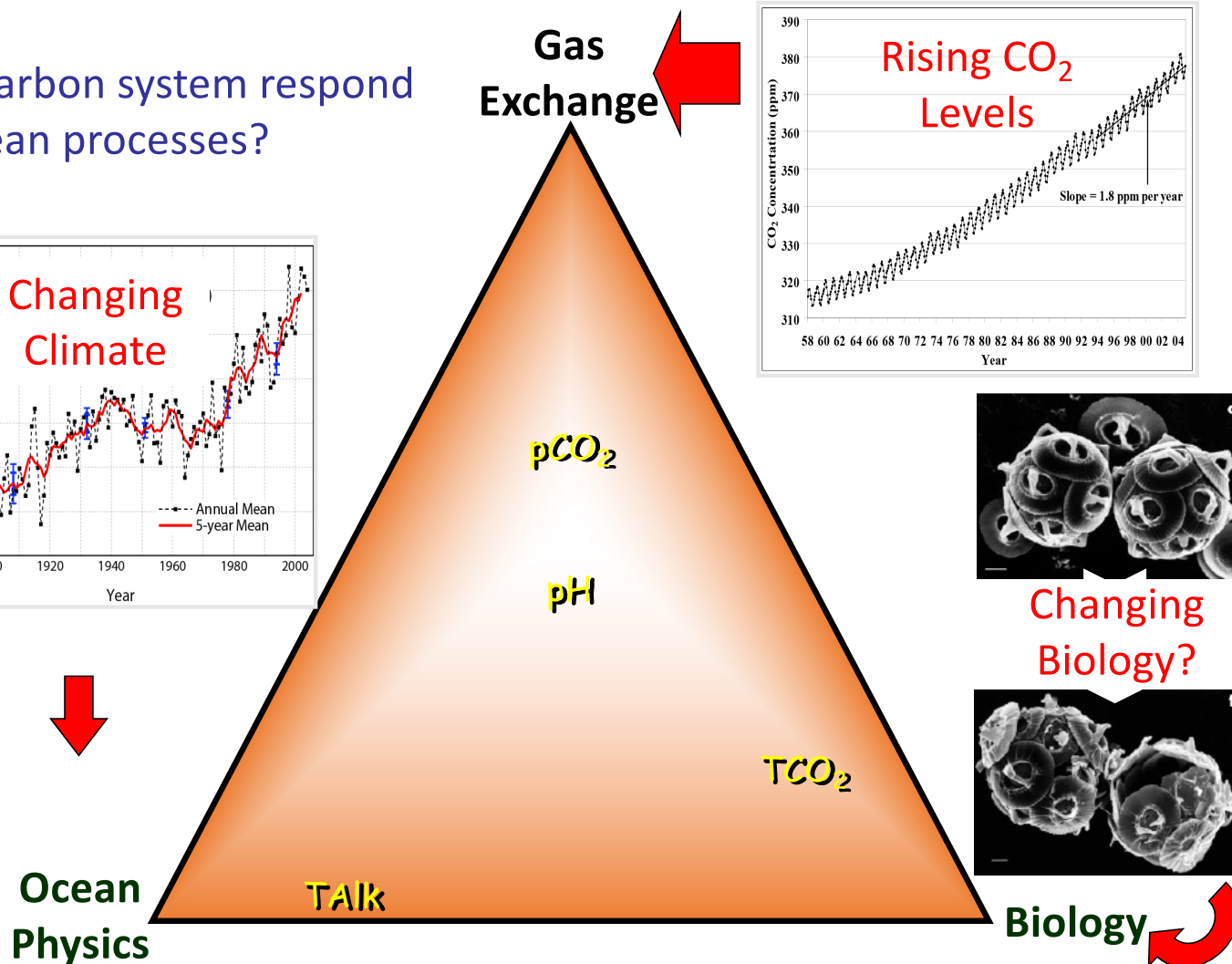
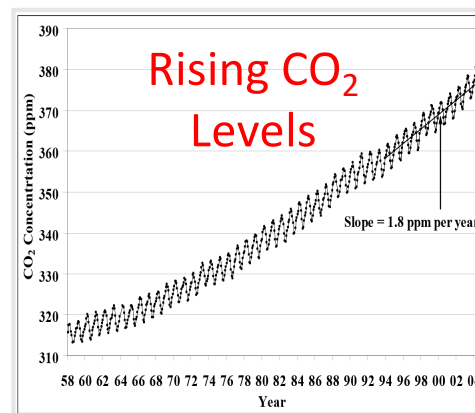
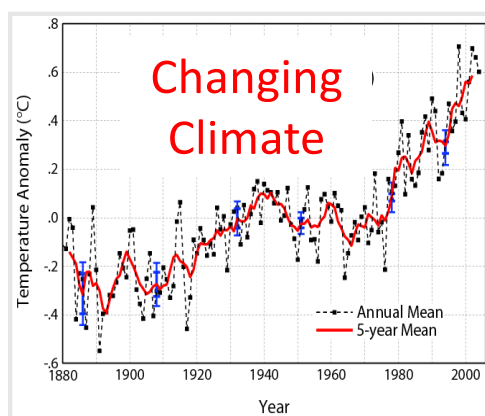
- 4 equations, 6 unknowns (H⁺, HCO₃⁻, CO₂, CO₃²⁻, DIC, TA)
- Measuring 2 allows the system to be constrained



Take home message:
Each of the 4 measurable
carbon parameters can tell us
something different about the
basic processes operating in
the Ocean



Question:
How will ocean carbon system respond
to changes in ocean processes?

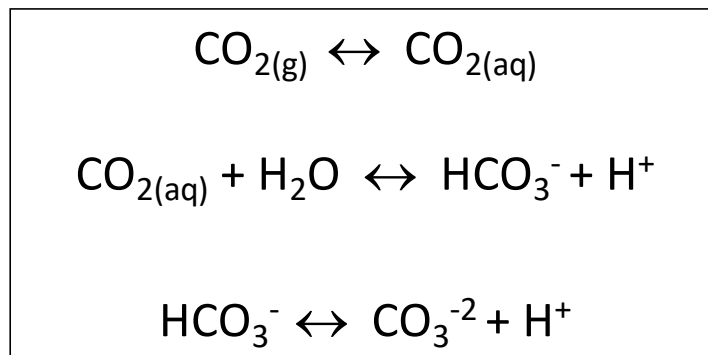


Changing
Biology?



Effects of Pressure on Carbonate Speciation

	<u>1 atm</u>	<u>1000 atm</u>
K_1^*	$10^{-5.89}$	$10^{-5.55}$
K_2^*	$10^{-9.13}$	$10^{-8.93}$



$$K_1^* = \frac{\{H^+\}[\text{HCO}_3^-]}{[\text{CO}_2]}$$

$$K_2^* = \frac{\{H^+\}[\text{CO}_3^{2-}]}{[\text{HCO}_3^-]}$$

As you raise a sample from depth:

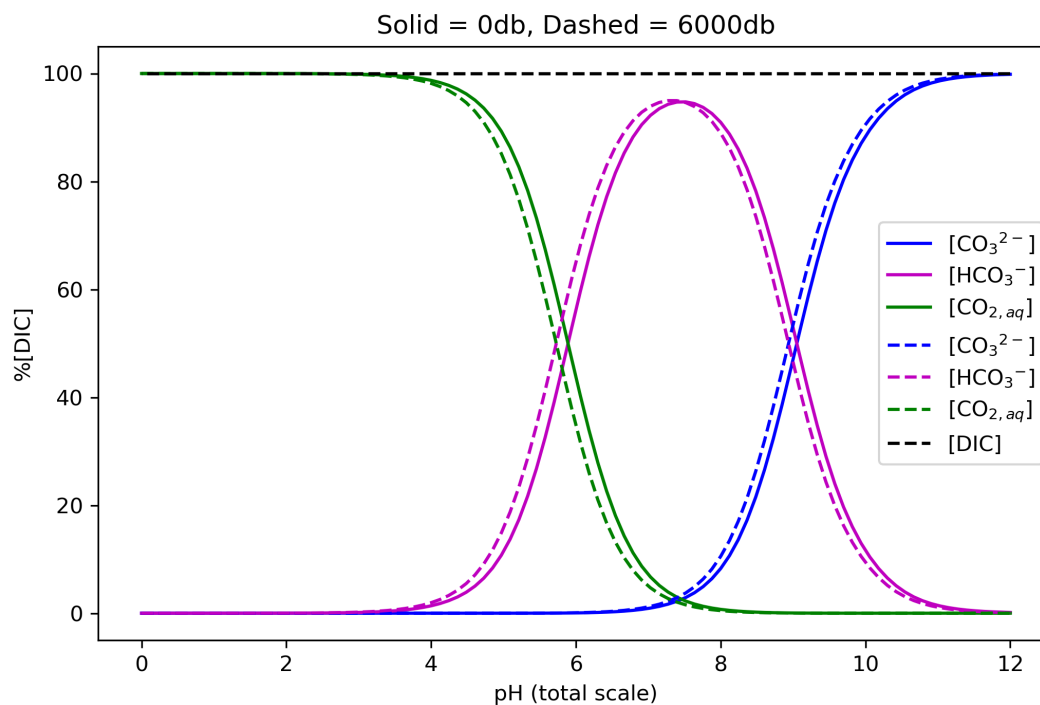
- K_s' decrease
- Reactions shift to left
- pH increases

Effects of Pressure on Carbonate Speciation

	<u>1 atm</u>	<u>1000 atm</u>
K_1^*	$10^{-5.89}$	$10^{-5.55}$
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As you raise a sample from depth:

- K_s' decrease
- Reactions shift to left
- pH increases



Chemical Reactions of Carbonate Species in Seawater

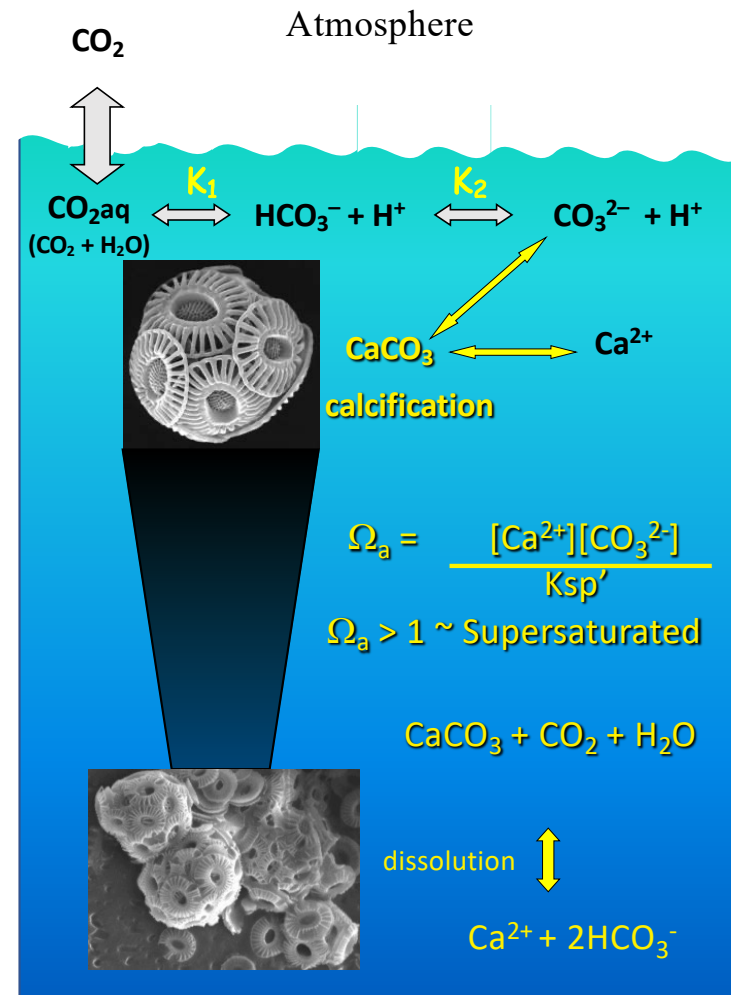
[Ca²⁺] is one of the 6 major ions in seawater

To a first order it is considered conservative with salinity (~10.3 mmol/kg at salinity of 35)

Therefore, [CO₃²⁻] primarily controls the saturation state of the waters with respect to aragonite, calcite, magnesian calcite

Solubility of CaCO₃ increases with decreasing temperature and increasing pressure.

Increasing Solubility from left to right:
calcite < aragonite < high-magnesian calcite

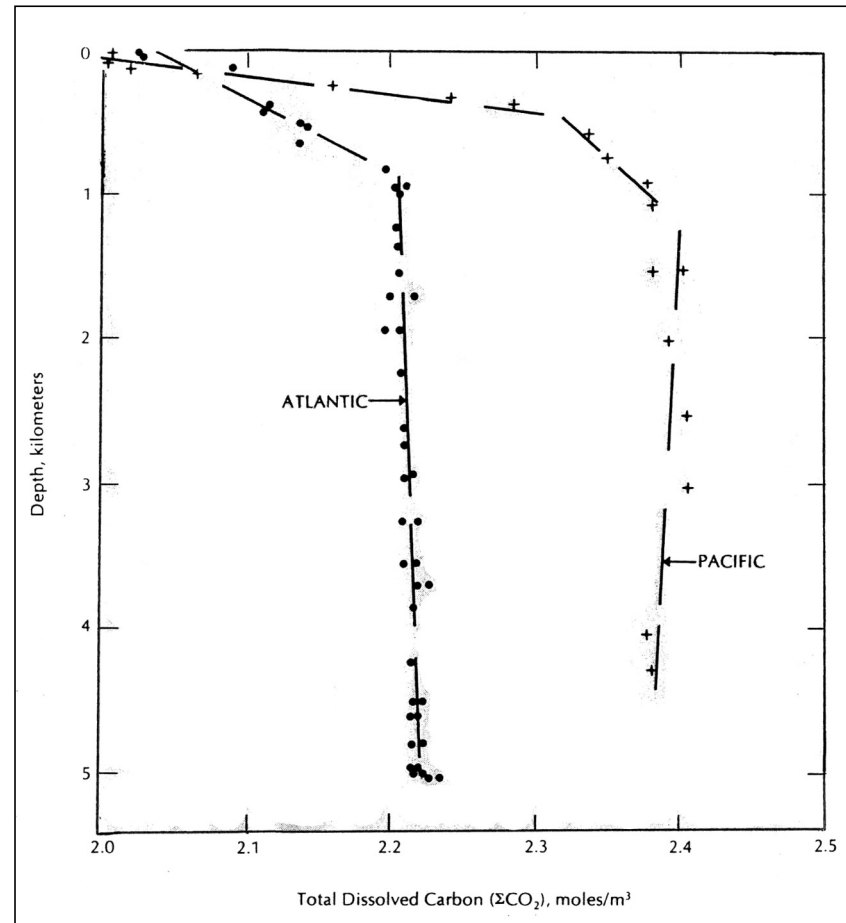


Calcite dissolution

- Remineralization of organic matter in the water column produces CO_2 that reacts with CO_3^{2-}



- Higher ΣCO_2 at depth
- Lower CO_3^{2-} at depth
- In the deep ocean, the decrease in $[\text{CO}_3^{2-}]$ from this reaction has a marked effect on CaCO_3 solubility



Saturation state of SW

- The saturation state of SW with respect to calcite (and aragonite, Mg-rich calcite, etc.) determines whether these phases will dissolve or not
- Define the *saturation state of SW with respect to calcite* (CaCO_3) as:

$$\Omega = \frac{(\{\text{Ca}^{2+}\}\{\text{CO}_3^{2-}\})_{\text{Seawater}}}{(\{\text{Ca}^{2+}\}\{\text{CO}_3^{2-}\})_{\text{Calcite-saturated seawater}}}$$

Assumes $\{\text{CaCO}_3\} = 1$

The denominator is also K_{sp} , the “solubility product”

- Seawater: Changes in $[\text{Ca}^{2+}]$ are small $\rightarrow$ changes in Ω largely controlled by $\Delta[\text{CO}_3^{2-}]$

Saturation state of SW

$$\Omega = \frac{\text{IAP}}{K_{\text{sp}}}$$

- **IAP= Ion activity product**
- Depth at which $\Omega = 1$ is called the **saturation horizon**
- **Where** $\Omega > 1$ water is supersaturated with respect to the ion concentrations therefore CaCO_3 will precipitate
- **Where** $\Omega < 1$ water is undersaturated with respect to the ion concentrations therefore CaCO_3 will dissolve
- K_{sp} dependant on T, P and mineralogy

Solubility of mixed carbonates

A variety of mixed carbonates, $\text{Ca}_{(1-x)}\text{Mg}_{(x)}\text{CO}_3$:

$$x = 0$$

Calcite or aragonite

$$x < 0.04$$

Low Mg-calcite

$$0.04 < x < 0.20$$

High Mg-calcite

$$x = 0.5$$

Dolomite

$$x = 1$$

Magnesite

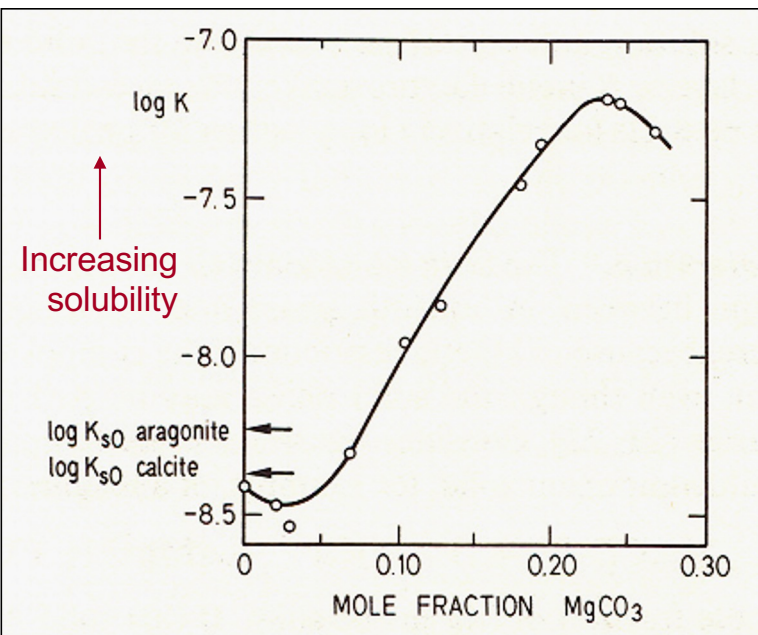
Solubility of Mg-calcite increases with increased Mg content



$$K = \{\text{Ca}^{2+}\} \{\text{CO}_3^{2-}\} / 1$$

Higher K means CaCO_3 dissolves with higher activities of Ca^{2+} and CO_3^{2-}

$\therefore \text{CaCO}_3$ is more soluble



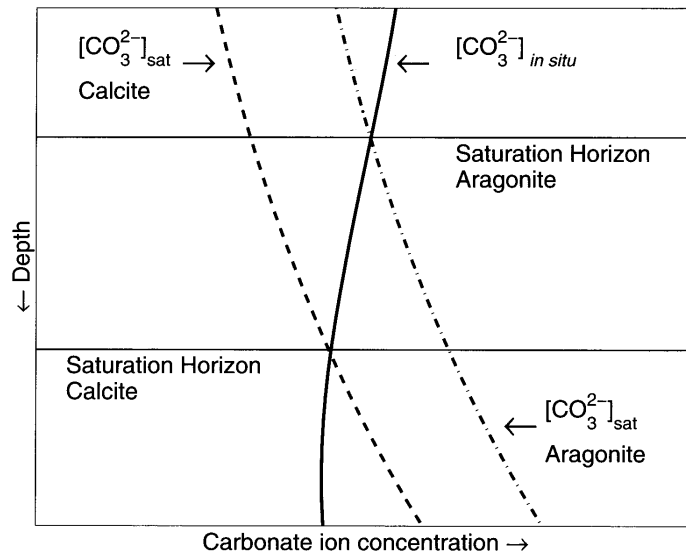


Figure 1.1.6: Illustration of the calcite and aragonite saturation horizon (SH) in the ocean (after Broecker and Peng, 1987). As pressure increases with depth, the solubility of calcite and aragonite increases as well ($[\text{CO}_3^{2-}]_{\text{sat}}$). The crossover between the *in situ* carbonate ion concentration (solid curve) and the saturation concentration for calcite (dashed curve) and aragonite (dot-dashed curve) determines the saturation horizon of the different mineral phases.

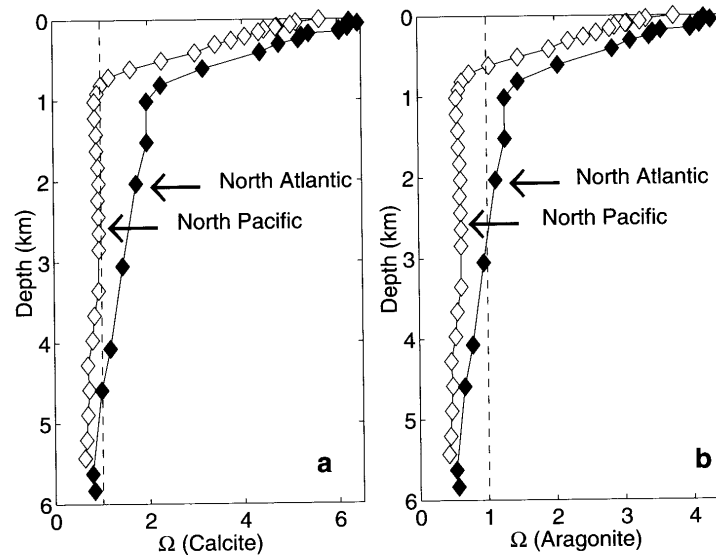


Figure 1.1.7: Saturation state of seawater, Ω , with respect to calcite (a) and aragonite (b) as a function of depth. Ω was calculated using Eq. (1.1.25) and DIC-TA profiles from the North Pacific and North Atlantic Ocean. The dashed vertical line separates areas of supersaturation ($\Omega > 1$) from undersaturation ($\Omega < 1$). Data are from WOCE Section P14N, Stn. 70 (F. Millero and Ch. Winn) and WOCE Section A05, Stn. 84, (F. Millero and S. Fiol), <http://cdiac.esd.ornl.gov/oceans/CDIACmap.html>.

Zeebe and Wolf-Gladrow (2001)

- The upper ocean is supersaturated wrt both calcite and aragonite
- Aragonite is more soluble than calcite in SW
- Saturation horizon is deeper in Atlantic than in Pacific

Carbonate Compensation Depth / Lysocline

- **Saturation Horizon** is where “ Ω ” = 1
- **Lysocline** is where dissolution effects first appear in carbonate grains
 - Since degree of saturation decreases with depth, dissolution rates should increase with depth
- **Carbonate Compensation Depth (CCD)** is where the depositional rate of carbonate is equal to the dissolution rate (i.e., no net accumulation of carbonates on the seafloor)
- The lysocline occurs above the CCD, but is at or below the saturation horizon because of
 - Kinetic effects
 - Protection by organic matter on particulates
 - Inhibitory effect of dissolved species like phosphate which have mid-depth maxima

